

ENERGYINTEL SERVICES LTD

GEOLOGICAL – GEOTECHNICAL INVESTIGATION AT THE SITE OF THE PROPOSED FOTOVOLTAIC PARK IN PLOT 316, SHEET/PLAN 39/47 SHA NICOSIA



SEPTEMBER, 2025

GEOINVEST LTD - ΓΕΩΕΠΕΥΝΑ

***Applied Geology – Geotechnics – Materials Testing
Environmental Engineering***

Viotechniki Periochi Aglantzias No.10, P.O.Box 20476, 2152 Aglantzia,
Tel: 22 33 00 93, Fax : 22 33 01 18, E-mail : geoinvest@cytanet.com.cy
Web: <http://www.geoinvest.com.cy>



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15/04/2025

X\Geotech2025\ FotovoltaicaEnergyintel316Sha

ENERGYINTEL SERVICES LTD,
Att.: Mr Marios Constantinou,
Nicosia.

Dear sir,

**SUBJECT: GEOLOGICAL – GEOTECHNICAL INVESTIGATION AT THE
SITE OF THE PROPOSED FOTOVOLTAIC PARK IN PLOT 316,
SHEET/PLAN 39/47 SHA, NICOSIA**

We are glad to advise you that the above ground investigation is completed.

The investigations comprised excavation of Trial Pits, in situ and laboratory testing, as well as the evaluation of the data obtained. A detailed account on the results of the ground investigation, are presented in the report that follows. It is believed that this work will facilitate the effort of your engineers in finding the most suitable solution to this case.

Finally, we would like to express our sincere thanks for entrusting our firm the execution of this site investigation and remain at your disposal for any clarifications or further information on this subject.

Sincerely yours,



GEOINVEST
ISO/IEC 17025:2017

Andreas Shathas
(Geologist - Managing Director)
Registration Number ETEK A035735

PRELIMINARY REPORT – EXECUTIVE SUMMARY

Geoinvest Ltd has been retained by Energyintel Services Ltd to conduct geotechnical field investigations and the related reporting including factual and interpretative report to support the foundation design of the structures of a photovoltaic park, in Sha area, Nicosia.

The park is planned to be developed in plot 316. The location of the site is shown in figures 1 and 2.

The field investigations were carried in September 9, in accordance with accepted standards, i.e. the Eurocode 7, CYS EN ISO 22476, CEN ISO/TS 17892, BS EN 5930 and BS EN 1377.

In summary, the purpose of the investigation according to the Client's requirements, is to establish:

- the general stratigraphy,
- the geotechnical properties of the strata underlying the site in relation to foundation design for the structures based on existing information, and
- to assess the excavation conditions so that the proper foundation type is applied.

The Scope of the investigation is:

- Site visit and geological survey
- Desk study based on existing geological and geotechnical information concentrated by Geoinvest Ltd and governmental data.
- Factual and Interpretative reporting, where the general and specific geological picture, the geotechnical parameters, the foundation and excavation conditions are, on a preliminary basis, evaluated.

All field and office works were plant and agreed with the client's representative.

The results of the investigations carried out can be summarized as below:

Morphology: The broader area is generally mildly hilly with small, rounded hills and with small, narrow, shallow streams. There is a general dip to the south. At the site under study, Plot No 11 is flat with a gentle dip to the east.

No cultivations or other commercial activities are practiced in the plot, except seasonal barley.

General Geology: The area is found within a "Lower Pillow Lava country" at the easternmost part of Troodos Ophiolite Complex. Close to the site, at its southwestern part a horst bringing the older Basal Group above the eruptive lavas, representing the older lava horizon mixed with dykes feeders to the younger two lava horizons, is a pronounce geological and morphological area feature.

Site geology and geotechnical conditions at the solar park site: From the geological point of view the materials encountered in the whole of the investigated sites are represented by Lower Pillow Lavas. From the geotechnical engineering point of view, they can be grouped into two Units:

- A. The Superficial soil representing the residual soils derived from the decomposition and disintegration of the underlying lavas.
- B. The in-situ Lavas, representing the bedrock of the area.

A gradation between the two units can be observed throughout the site.

Geotechnical conditions at the solar park site: About 35% of the plot area is covered with residual soil (the cultivated part, as shown on the google earth map) and along the rest of the area the lava-bedrock is outcropping on the surface.

The superficial-residual soil is of a few cm to more than 2 meter thick in most of the cultivated plot area and consists of variably shaped and sized, angular lava fragments surrounded by fine material of the same origin of mostly sand and less silt and clay fractions. The lava fragments are of fine gravel to cobble size, occasionally of boulder size.

The lavas are generally weathered, altered and discolored to brown, yellowish and reddish brown colors.

Hydrogeology: No groundwater is envisaged at shallow depth. The permeability of the Fill is high and that of the bedrock medium on top and then low.

The electrical resistivity is high (mostly $>377 \Omega m$) due to the rocky character of both the Soil and the Lavas. It varies depending on grains size, density, weathering, and moisture content. The results show that the more the gravel fraction and the lower the weathering of lavas the higher the electrical resistivity.

The soil corrosivity is negligible/low to moderate.

The Thermal Conductivity is variable, but within the expected limits, depending on various factors, like moisture content, density, mineral and chemical composition of the soils, soil structure and grain size.

Archaeology: No signs of archaeological monuments or remnants were observed during the walkover of the site.

Foundation Conditions: Taking into consideration the type of structures to be erected, the site offers very good foundation conditions with allowable bearing capacities much higher than the loads to be imposed on the ground.

Foundation type: All types of foundation, shallow pad footings or piles (partially driven and partially predrilled), could be applied for this case. Whatever the choice of foundation type is, it should be placed to such a depth (≥ 1.0 m in the case of piles and ≥ 0.5 m in the case of pad foundation), so that they are protected from overturning due to lateral wind loads. The final foundation depth and dimensions should be decided by the engineers of the project taking into consideration the load of the structures as well as the external loads and moments acting on the structures and the foundations. Since the loads of the structures are very small, no notable total or differential settlements are expected (<10 mm). Furthermore, the settlements will be immediate due to the granular nature of the sub-ground.

The roads could be constructed considering CBR of the order of $>11\%$.

Excavation conditions: If the proper machinery is used, no issues with the excavations and site cleaning and levelling, if necessary, are envisaged. The excavation conditions, from the geotechnical point of view, are classified as "soft/easy". Even the small, backhoe excavators, 8-10 tons (~ 100 HP) could be suitable for small scale excavations or leveling of the site. But for mass excavations excavators of at least 25- 45 tons might be necessary.

The main **geological hazards** that could be considered in the area are:

The moderate seismic risk. In accordance with the National Annex of Eurocode CYS EN 1998-1: 2004, the area is found within seismic zone 2 with peak ground acceleration 0.20. Based on table 3.1 of the Eurocode, the soils/rocks in the investigated area are variable and could be classified as Ground Type A (lavas) and C, (Superficial soil).

No high flood risk is envisaged, but an effective drainage system should be designed so that surface erosion and infiltration of storm water into the ground are avoided. Infiltration of storm water into the ground and the alternating wet and dry conditions may provoke loosening and settlement of the soils with negative effects on the structures' foundation. An effective storm water management, therefore, could minimize this risk.

Active faults, Flood risk, Karst risk, Liquefaction: No such risks are considered.

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1 INTRODUCTION

Energyintel Services Ltd retained GEOINVEST LTD to conduct geotechnical field investigations and the related reporting including factual and interpretative report to support the foundation design of the structures of a photovoltaic park, in Sha Nicosia area. The location and the plot of interest are shown in figure 1. It concerns the area covered by the parcel No 316, Sheet/Plan 39/47.



Figure 1: Site Location

1.1 Brief description of the proposed construction

The development consists of a PV park in plot 316, Sheet/Plan 39/47 of total area 29767 m², in Sha, Nicosia area.

1.2 Purpose and scope of work

This technical engineering geological report intends to assimilate data from the ground investigation, which included desk study, site geological survey, presentation of the works and geological evaluation of the results in a factual and interpretative report.

The investigations were carried in September 2025, in accordance with accepted standards, i.e. the Eurocode 7, CYS EN ISO 22476, CEN ISO/TS 17892, BS EN 5930 and BS EN 1377 and included:

- desk study,
- excavation of trial pits,
- soils sampling and chemical analyses to establish soil acidity and in combination with Soil Resistivity and Thermal Conductivity to establish soil corrosivity so that the proper metal finishing of buried ferrous metals is chosen and the proper concrete to be used.
- laboratory testing to establish the particle size distribution and plasticity characteristics, particle density and bulk density.
- in situ dynamic testing to assess foundation conditions,
- presentation of the works and engineering geological evaluation of the results in a factual and interpretative report.
- Assess the excavation conditions and recommend erosion prevention measures.

In summary, the purpose of the investigation according to the Client's requirements, included the following:

- The general stratigraphy of the site,
- The geotechnical properties of the strata underlying the site in relation to foundation design for the structures and
- assess the excavation conditions so that the proper foundation type is applied.
- Present the results on a Factual and Interpretative report, where the general and specific geological picture, the geotechnical parameters, the foundation, and excavation conditions are evaluated.

The results are presented in this report, which consists of one Volume.

1.3 Structure of report

This report is organized in the following sections:

FACTUAL PART OF THE REPORT

1. Introduction
2. A brief description of the site
3. Regional Geology of the area
4. Ground Investigations, works carried out

INTERPRETATIVE PART

5. Subsurface conditions and characteristic geotechnical parameters along the investigated site.
6. Foundation conditions at the structures sites - Geotechnical considerations for foundations.
7. Excavation conditions
8. Hydrogeological conditions
9. Archaeological Conditions
10. Geologic Hazards
11. Conclusions and Recommendations

1.4 Units of measure

In this report the International System of Units (SI) is used where length is described in meters (m), force in Newton (N), stresses and pressures in Pascal (Pa). For larger or smaller quantities, the appropriate prefix is used (e.g. mm, kN, kPa or MPa).

2 SITE DESCRIPTION

2.1 Site Location

The site of development is located about 1.2km to the south of Sha and 3,5 Km to the southwest of Mosphiloti communities, to the west of A1 highway (Nicosia Limassol motorway), as shown on the location plan on figure 1.

2.2 Site Morphology

The broader area is generally mildly hilly with small, rounded hills and with small, narrow, shallow streams, as shown on the attached figures and photos. There is a general dip to the east. At the site under study, Plot No 316 is slightly undulating. A few fields present used to be cultivated for livestock production.

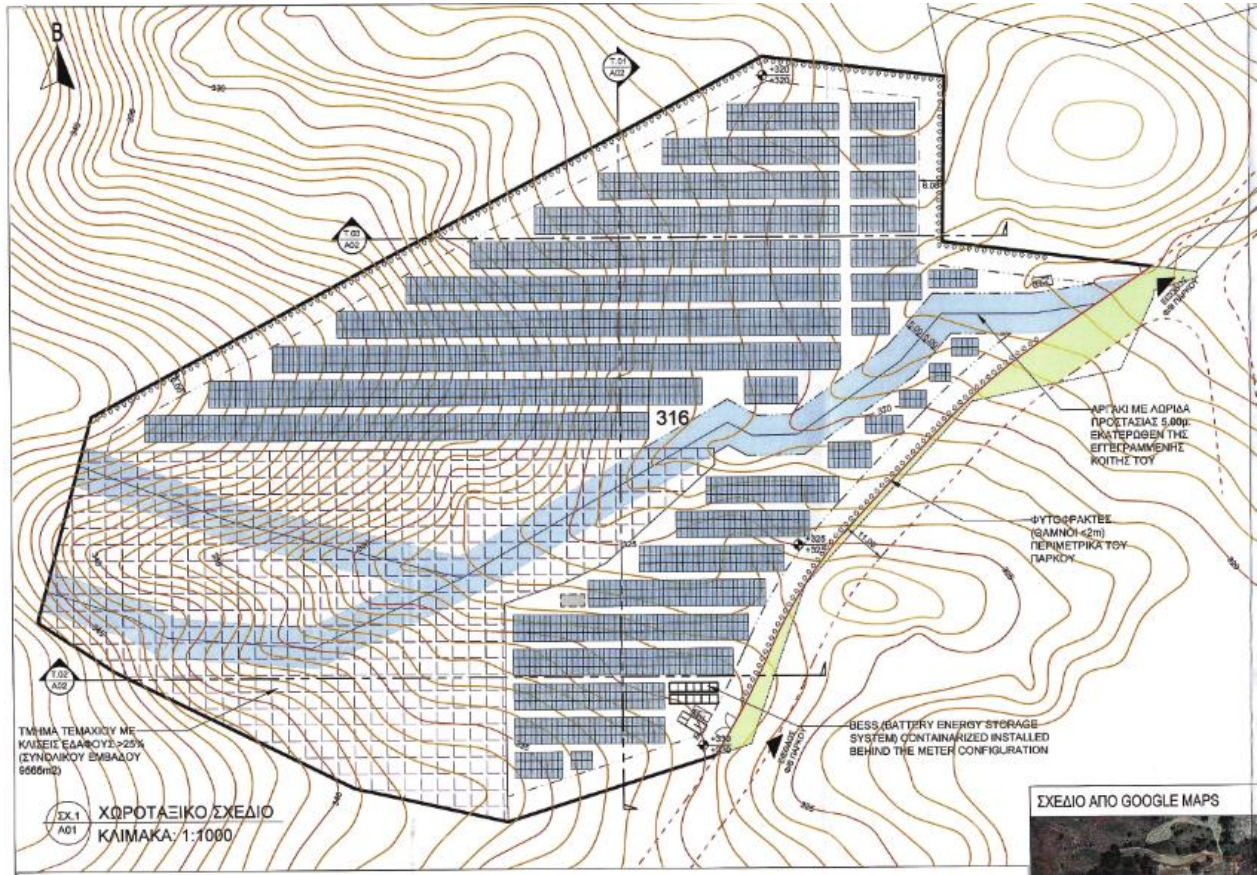


Figure 2. Topographical Map of the plot and lay out of the panels

3 AREA GEOLOGY AND STRATIGRAPHY

The broader area is represented by the following geological formations (from the older to the younger):

3.1 Igneous rocks of “Troodos Ophiolite Complex”

The general geology is represented by the The Eruptive Sequence, consisting of

- The Basal Group
- The Lower Pillow Lava and
- The Upper Pillow Lava.

Due to the high degree of weathering, decomposition and disintegration residual soils have been developed and colluvium deposited on top of the above rocks and along the slopes. Furthermore, along valleys and streams, recent, alluvial, valley and reworked colluvial material was accumulated.

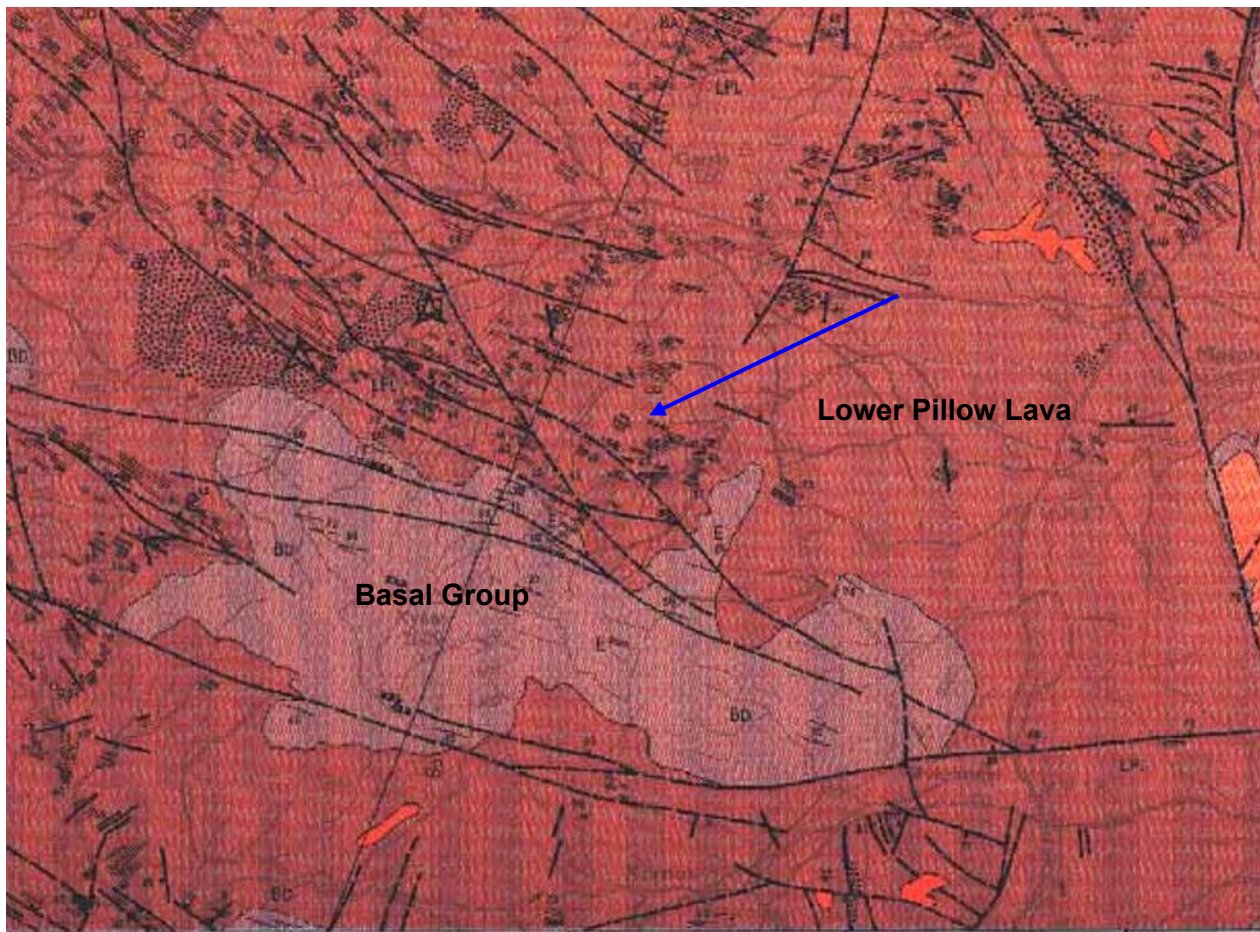


Figure 3. General Geological Map from Geological Map of “Dali Area” (GSD)

3.2 Regional structural setting

As described at the beginning of this chapter, Cyprus Island is located in an active and complex geodynamic setting, which is the result of different tectonic phases, which, in accordance with recent studies, led to the following fault systems:

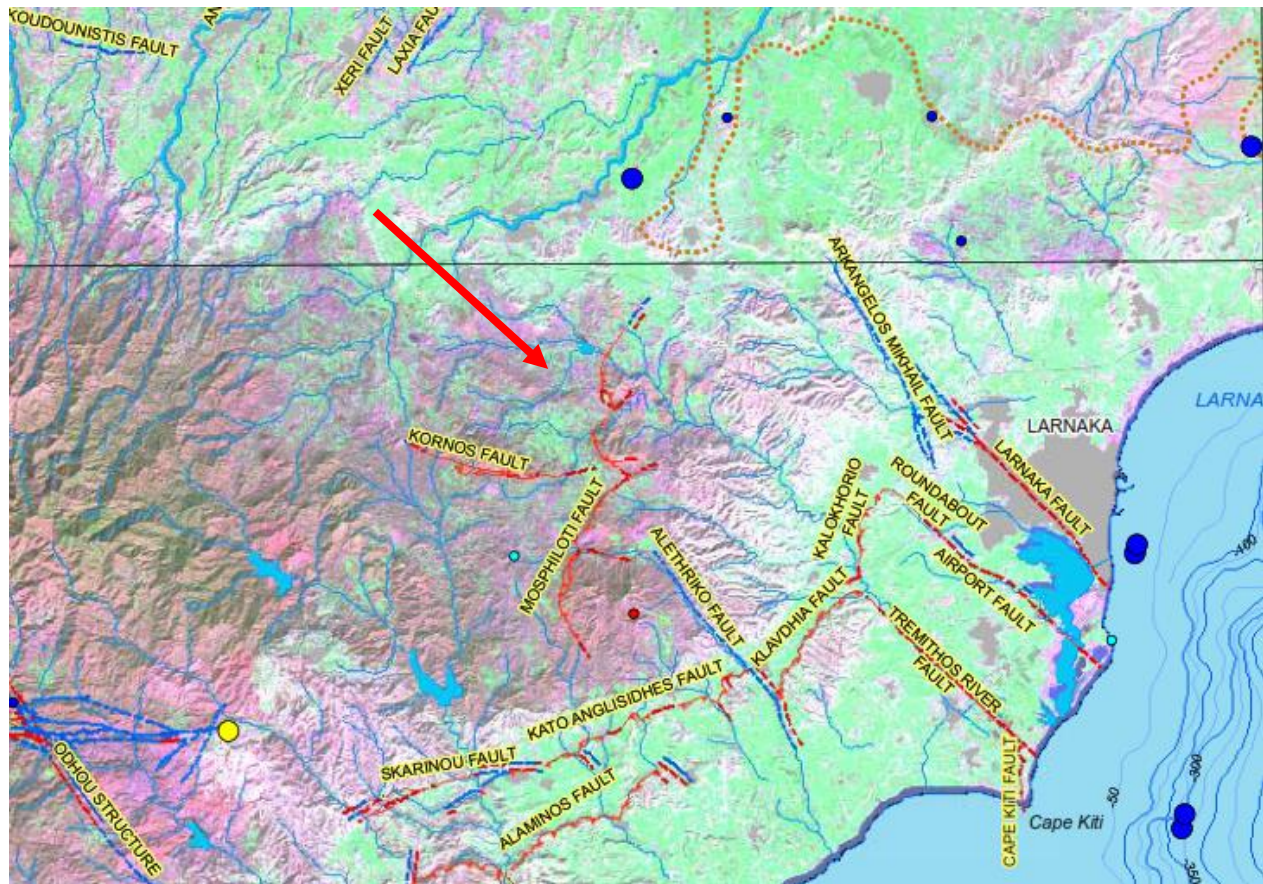
- E-W extension in west and southwest Cyprus, during **late Miocene extension**, led to the formation of faults of N-S orientation, whereas in south central Cyprus, NW-SE extension led to the formation of NE-SW faults
- The localized NNE-SSW extension during the **late Pliocene** led to the formation of WNW-ESE faults
- A pervasive compressional/transpressional event, which occurred throughout the south of Cyprus during **Pleistocene-Recent compression and transpression** led to the reactivation of pre-existing structures, and the formation of E-W and NNE-SSW-trending left lateral faults and NNW-SSE-trending right lateral faults

3.2.1 Local structural setting and seismicity

With regards to local structural setting, the area under study is found within Maroni-Psematismenos Basin. In accordance with Bear (1960) and Pantazis (1967a, b) three populations of faults were identified:

- N330-N350 (NNW-SSE) striking faults;
- N070-N080 (ENEWSW) striking faults; and
- N010-N040 (NNE-SSW to NE-SW) striking faults.

Several NNW-SSE striking, sub-vertical strike-slip faults have also been identified in the axis of the Maroni Basin by Soulas, 2003. In addition to the NNW-SSE-striking fault set, the Maroni Basin is dissected by several ENE-WSW-striking, SSE-dipping normal faults. In the central axis of the basin, near Maroni village, Kalavasos Formation evaporites are cut by a conjugate set of fractures aligned N340-N020 (N-S) and N280-N300 (WNW-ESE). From the Cyprus Geological Survey Department data, Kinnard (2008) compiled from GIS fault populations the structural geological scheme of Maroni-Psematismenos Basin, as shown on figure 3.



Legend

Faults

Quaternary faults

- Fault with Quaternary displacement
- - - Inferred fault with Quaternary displacement
- Older fault with Quaternary displacement
- - - Older fault with inferred Quaternary displacement
- - - - Inferred fault with inferred Quaternary displacement
- Fault scarp, scarplet
- Faulting within Plio quaternary deposits
- Reverse fault
- Frontal evidence of uplift, produced by blind thrusting

Other tectonic structures

Lineation

- Lineation with inferred faulting
- - - Inferred lineation with inferred faulting

Fold

- Anticline

Block tilting

- Tilting

Gravitational structures

- Gravitational pseudo-fault
- - - Inferred gravitational pseudo-fault
- Landslide crest

Other faults

- Older fault concealed by younger deposits
- - - Fault with Quaternary displacement, concealed by younger deposits
- Fault zone
- - - Inferred older fault with Quaternary displacement, concealed by younger deposits
- - - Inferred fault with Quaternary displacement, concealed by younger deposits
- - - Inferred old fault, concealed by younger deposits
- Older thrust

SISMICITY (period -2150 / 2005)

Depth (Km)

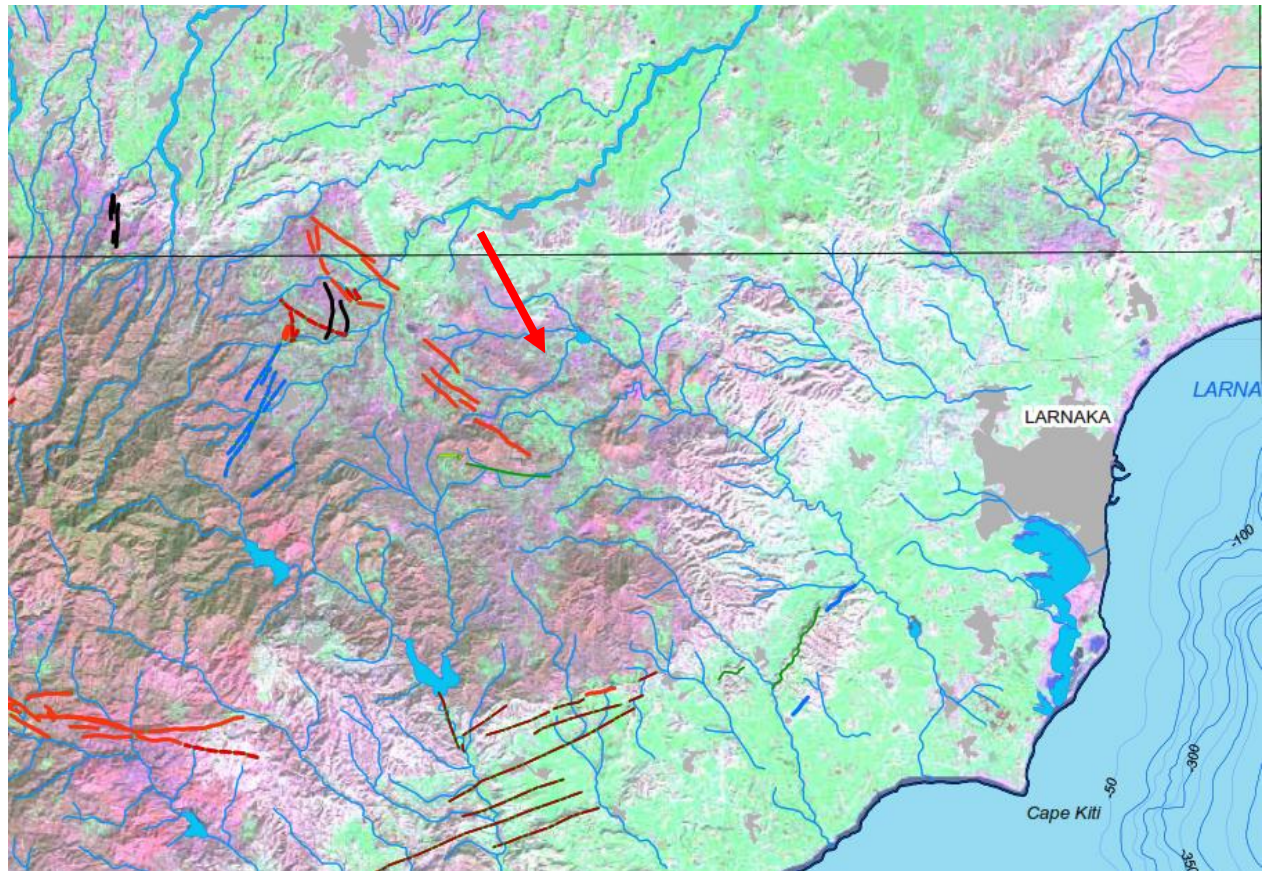
- < 10
- 10.1 - 20
- 20.1 - 30
- > 30

Magnitude

- 3.0 - 4.0
- 4.1 - 5.0
- 5.1 - 6.0
- 6.1 - 7.0

- - - Survey limits

Figure 4. Quarternary faults Limassol area, Geoter, Neocyp Consortium & B.Grellet, «Study of active tectonics in Cyprus for seismic risk mitigation



Fault

- | | |
|-----|-----------------------------------|
| --- | Faults identified presumably |
| — | Faults identified with confidence |
-
- Active faults
 - Presumably active faults
 - Potential signs of surface ruptures and secondary ground deformations
 - Lineaments
 - Lineaments identified on CORONA images in the shelf area
 - Potential faults identified on bathimetric map

Figure 5. Part of the map of The Active Faults in Cyprus, Geoter, Neocyp Consortium και B. Grellet, «Study of active tectonics in Cyprus for seismic risk mitigation, WP8-Final Report» Nicosia, Cyprus, 2005

4 GROUND INVESTIGATIONS – WORK CARRIED OUT

4.1 Standards and Regulatory Framework

The following reference standards were considered for the geotechnical survey:

- Eurocode EC 7 Geotechnical design - part 1 General Rules
- Eurocode EC 7 Geotechnical design - part 2 Ground investigation and testing
- CYS EN ISO 14688-1:2002+A1:2013 -- Geotechnical investigation and testing -- Identification and classification of soil -- Part 1: Identification and description;
- CYS EN ISO 14688-2:2002 Geotechnical investigation and testing -- Identification and classification of soil -- Part 2: Principles for a classification
- ISO 22476-3:2005+A1:2011 Geotechnical investigation and testing -- Field testing -- Part 3: Standard penetration test
- CYS EN ISO 14689-1:2003 Geotechnical investigation and testing -- Identification and classification of rock - Part 1: Identification and description

4.2 Field and Laboratory Works

A walkover helped to organize the Intrusive investigation by means of

- Excavation, description, photo taking of trial pits,
- In situ testing using a Dynamic Cone Penetrometer,
- Sampling,
- Laboratory testing
- Site reinstatement.

The **excavation of trial pits** was performed using a backhoe excavator.

In situ testing by means of Dynamic Cone Penetration Tests (DCP) were performed during the trial pits excavation and sampling campaign to establish the soil density and hence the bearing capacity of the ground. The DCP test measures the resistance offered to the penetration of a standard closed cone, which can be empirically correlated to soil density, strength, and stiffness. The results were converted to SPT results from which the allowable bearing pressures could be calculated.

Sampling was performed by means of bulk samples representing the whole thickness of the surface soil on top of the membrane and placed in sound plastic containers. All of them were properly wrapped and taken to the Laboratories for testing. Wrapping was performed using a very thin, plastic, tough membrane, which can be stretched and tightly seal the sample, so that the latter is protected from drying out. A total of 30 samples were taken from the site.

Resistivity and Thermal Conductivity were carried out in the laboratory in several types of samples. The location of samples tested are shown in figure 6. The locations were chosen in such a way so that a representative picture of the whole site is obtained. The equipment and methodology used are presented in Appendix 1. The results were used in combination with the chemical analyses so that the soil corrosivity is established.

The Laboratory tests for the physical properties of the soils were performed in accordance with the relevant European CYS CEN ISO/TS 17892:2004 series, BS EN 1377-3:1990 and BS EN 13925-1:2003 as shown on the following table and involve identification tests, i.e.

particle size distribution by means of

- wet sieving,
- bulk and dry density,
- specific gravity
- natural moisture content,
- Shear tests in small shearbox,
- Atterberg Limits and Linear Shrinkage.

Selected samples passed the necessary treatment and preparation and then sent to the chemical laboratory. The chemical analyses included the following parameters:

Sulphates (SO₄),
Chlor ions (Cl⁻),
Redox Potential Mv,
pH.

A short description of the tests carried out with reference to the Standards followed is presented below.

Natural Moisture Content

The natural moisture content in accordance with CYS CEN ISO/TS 17892-1:2004, was determined for all samples tested for Electrical Resistivity and Thermal Conductivity.

Particle Size Distribution

Combined wet sieving/hydrometer analyses were carried out in accordance with the requirements of CYS CEN ISO/TS 17892-4:2004. The dispersing agent for the hydrometer was sodium hexametaphosphate and the sample used was oven dried.

Atterberg Limits

Testing was carried out in accordance with the requirements CYS CEN ISO/TS 17892-12:2004. The Cone penetration apparatus (method 2B) was adopted for the liquid limit tests. The results are presented graphically on individual test sheets and on the Casagrande Plasticity Classification Chart. All samples tested for Atterberg Limits were also tested for linear shrinkage.

Direct Shear Strength test

The direct Shear Strength tests have been carried out in accordance with the CEN ISO/TS 17892-10:2004. The load used in each specimen was calculated with relation to its overburden and the shearing was applied slowly enough to allow excess pore pressures

to dissipate by drainage so that effective stresses are equal to total stresses. Thus, the effective shear strength (cohesion) and the effective angle of internal friction of soils can be determined.

Table 1: Laboratory tests - relevant standards

Classification Tests	<i>Particle Size Distribution by means of both wet sieving and hydrometer</i>	CYS CEN ISO/TS 17892-4:2004
	<i>Bulk and Dry Density</i>	CYS CEN ISO/TS 17892-2:2004
	<i>Natural Moisture Content</i>	CYS CEN ISO/TS 17892-1:2004
	<i>Atterberg Limits</i>	CYS CEN ISO/TS 17892-12:2004
	<i>Linear Shrinkage</i>	CYS CEN ISO/TS 17892-12:2004
	<i>Specific Gravity/Particle Density</i>	CYS CEN ISO/TS 17892-3:2004
Strength tests	<i>Direct Shear Strength Tests</i>	CYS CEN ISO/TS 17892-10:2004
Electrochemical Tests for Soils	<i>Sulphates, Total sulfur, Chlorides, pH, Redox Potential,</i>	As per relevant tables
	<i>Electrical Resistivity and Thermal Conductivity</i>	As per Appendix 1

4.3 Desk study

Further to the data obtained during this investigation, a lot of information was obtained from several studies undertaken by Geoinvest in the area around.

4.4. Data Evaluation, Report

All the data gathered from the field and from the above studies have been evaluated and interpreted by the specialists of Geoinvest Ltd.

In the following chapters, the report describes all the works and methodology followed and presents the results on which all the conclusions and recommendations have been based on. The results of both the field and desk studies were used to prepare the geological model, which helps to understand the general geology of the area and the geological relationship between the various soil and rock formations.

5 SUBSURFACE GEOLOGICAL AND GEOTECHNICAL CONDITIONS AT THE INVESTIGATED SITE

5.1 Geological model

The area is found within the Eruptive Sequence of the Troodos Ophiolite Complex, represented by all three Lava Horizons, which have undergone the zeolite and the lower greenschist facies of metamorphism. Due to the high degree of weathering, decomposition and disintegration of the uppermost part of lavas, residual soils have been developed and alluvial/colluvial material was accumulated along the slopes and stream valleys.

5.2 Site Geology and Geological relationships

From the geological point of view the materials encountered in the whole of the investigated sites are represented by lavas as described above. From the geotechnical engineering point of view, they can be grouped into two Units:

- A. The Superficial Soil and
- B. The in-situ Lower Pillow Lavas, representing the bedrock of the area.

Their relationship with the rest of the geological formations is shown on table 2 and on the geological map in figure 5.

Table 2: Stratigraphic relationship

Approximate Geological Age	Formal lithostratigraphic classification	General description
Quaternary-Recent (Pleistocene to Present!)	Man Made Ground	Made ground in areas of excavations and quarries
	Recent Alluvium, Colluvium, Top Soils, Residual soils	Superficial, Alluvial/Valley and Colluvial Deposits, mostly clays, silts and sands, occasionally gravel, all of sedimentary origin
Tertiary-Quaternary (Pliocene- lower Pleistocene)	Athalassa Formation	Calcarenes incorporating in places thinly to thickly laminated sandy marl
		Conglomerate and cemented gravel with sand, marl matrix
Tertiary (Pliocene)	Nicosia Formation	Massive to thickly laminated marl and sandy marl
Tertiary (Upper Miocene/ Messinian)	Pakhna/Kalavassos Formation	Gypsum bodies, Marl, sandy marl and chalky marl, locally gypsum bearing
Tertiary (Upper Miocene)	Pakhna/Koronia Limestone	Reef Limestone
Middle to Upper Miocene	Pakhna Upper Sequence	Limestone, Calcarenite Silty Sandstone, Marls, Sandy Limestones, Chalks
	Chalk and Marl sequence	Chalk and Marl
Middle Miocene	Pakhna Lower Sequence	Massive and Cleaved Chalk
	Lefkara Formation	Upper Marl, Chalk and Mar - Upper Chalks, Chalk and Chert Lower Marl
Upper Campanian/ Maestrichtian	Kannaviou/	Bentonitic Clay
Maestrichtian	Troodos Ophiolite Complex	Igneous rocks of Troodos Ophiolite Complex
<i>The green highlighted are present in the area under study or at its close vicinity.</i>		

5.3 Engineering Characteristics of the soils at the site under study

From the geological point of view the materials encountered in the whole of the investigated area are represented by Lavas covered in some parts by the Superficial, Residual Soils of variable thickness, from a few cm to a couple of meters, as shown on figure 7 below.

From the geotechnical engineering point of view, they can be grouped into two units, coinciding with

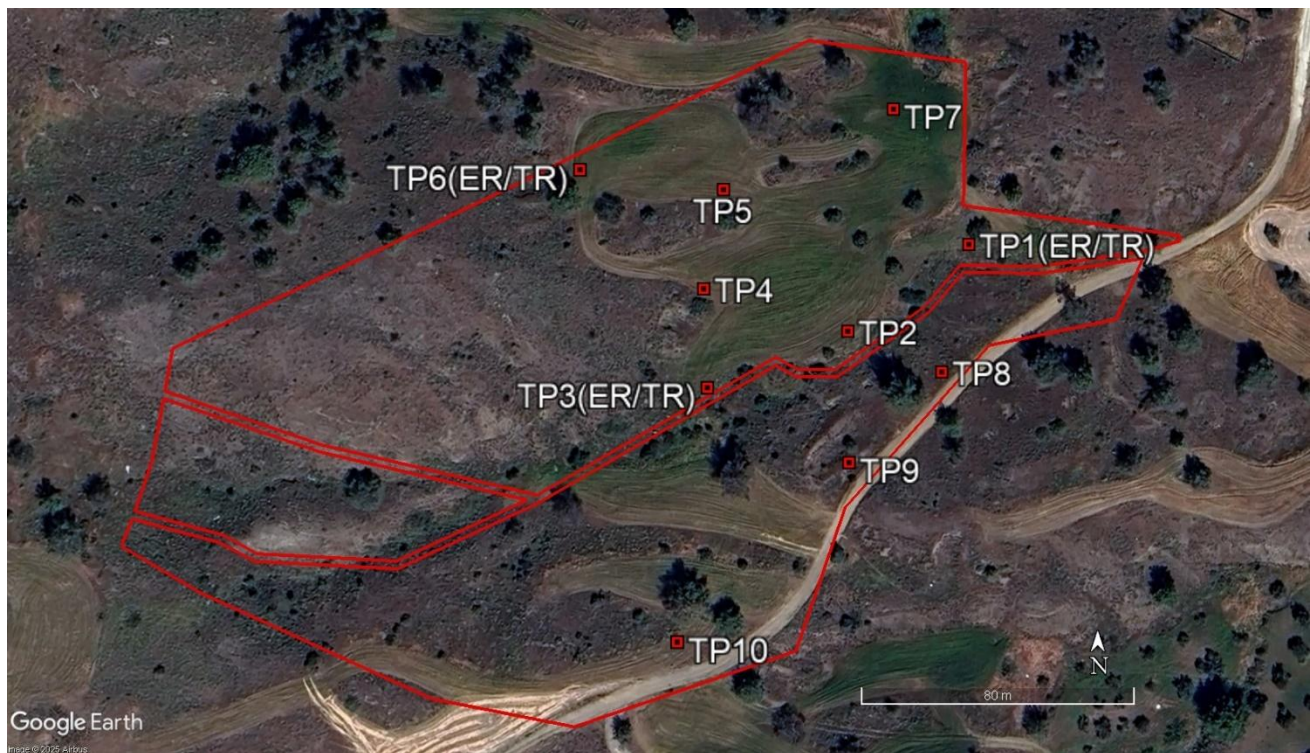
- **The Soil cover and**
- **the in-situ lavas.**

Their engineering parameters, classified into three main groups,

- index properties,
- strength/density properties and
- chemistry,

are presented in the following paragraphs.

The distribution of the two units is presented in the following figures and described in the following paragraphs.



Explanations: TP: Trial Pits, ER/TR: Electrical Resistivity, Thermal Conductivity and El. Resistivity (Laboratory)

Figure 6. Location of Field works

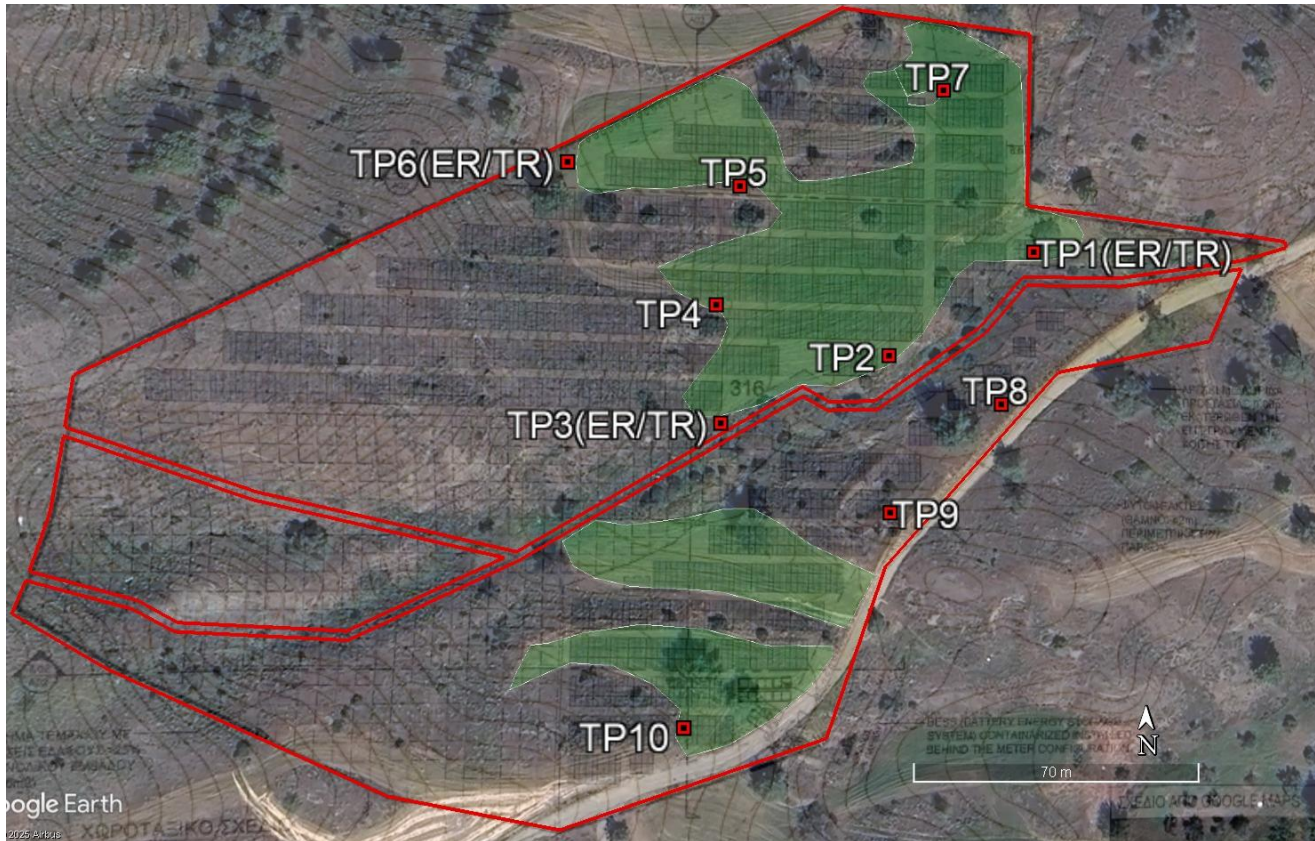


Figure 7. Lay out of the panels and areas with >1 m thick soil cover (green colored)

➤ **Unit A – Soil Cover**

The Soil cover, developed from the weathering, decomposition and disintegration of the lavas, is a few cm to more than 2 meters thick in about one third of the plot area. It consists of variably shaped and sized, angular lava fragments surrounded by fine material of the same origin of mostly sand and less silt and clay fractions. The lava fragments are of fine gravel to cobble size, occasionally of boulder size. The unit, in general, is of variable density.

Index Properties

The various soil fractions are represented in varying proportions as shown on table 3 below.

Table 3: Particle Size Distribution of Unit A

Terra Rosa Top Soil	Gravel %	Sand %	Silt %	Clay %
No of tests	3			
Minimum	37	15	6	2
Maximum	77	34	21	8
Average	57,0	25,3	13,3	4,3
Standard Deviation	20,0	9,6	7,5	3,2
COV%	35	38	56	74

PSD analyses were performed on material <cobble size, i.e. <60 mm. The content of the silt and clay fractions is related with material smaller than <60 mm and not with the whole of the material.

The Liquid Limits of the soil cover are of the order of 20 – 36 % and corresponding Plasticity Index of the order of 8 – 18 % depending on the content of clay. They can be classified as soils of low to intermediate plasticity/ compressibility on the Casagrande Plasticity Chart – Soil type CL/CI. The Linear shrinkage was found to be within the range of 5 – 10%.

Table 4: Plasticity Characteristics – Unit A

Terra Rosa Top Soil	LL %	PL %	PI %	LS %	Soil type	Moisture Content %
No of tests	3					
Minimum	20	12	8	5	CL/CI	4 – 10
Maximum	36	18	18	10		
Average	29,3	15,7	13,7	7,3		
Standard Deviation	8,3	3,2	5,1	2,5		
COV%	28	21	38	34		

The **Moisture Content** ranges between 4 to 10 %. The **Specific Gravity** is of the order of 2.215 – 2.355 g/cm³. The **Bulk Density** is of the order of 1.550 – 1.915 g/cm³ and the Unit Weight of the order of 15.2 to 18.8 KN/m³.

Density and Strength Parameters

Based on DCP tests results, converted to SPT, they can be characterized as loose and medium dense, with assessed SPT – N values, per 30 cm of penetration of the order of 9 to 25, with the majority, however, to be limited below 18. This test, unfortunately, is mostly influenced and masked by the presence of coarse gravel and cobble size rock fragments. Many results were recorded with high N-values or met with refusal due to the presence of the rock fragments. They were, therefore, treated with care and the masked ones were not considered for evaluation.

No strength tests by means of **Unconfined Compressive Strength (UCS)** or **Shear Strength** could be performed.

Direct Shearbox test have also been conducted and **the effective angle of internal friction** was found to be of the order of $29 - 34^{\circ}$ with effective cohesion of the order of $3 - 10$ kPa. Based on this test the undrained shear strength (strength peak values) were found to be of the order of $13 - 14$ kPa.

Table 5: Density, Unit Weight, Specific Gravity

SPT (Converted DCP)	Bulk Density g/cm ³	Unit Weight kN/m ³	Specific Gravity g/cm ³
9 – 25	1.550 – 1.915	15.2 – 18.8	2.215 – 2.355

Table 6: Strength, Cohesion, Friction Angle

UCS (q _u) kPa (From DCP/SPT)	Undrained Shear Strength (0.5 UCS) c kPa	Shear Box		
		Effective ϕ degrees	Effective Shear Strength c (drained) kPa	Undrained Shear Strength c (shearbox peak values) kPa
54 – 150	27 – 75	$29 - 34^{\circ}$	3 – 10	13 – 14

Soil Parameters – Ground Coefficients

Based on various methods, the Es of the superficial soils could be of the order of 20 MPa and the Coefficient of Subgrade Reaction of the order of 30 MPa.

Chemical characteristics

The pH is of the order of 7.8 – 8.2. The sulphates content (SO₄), are of the order of 0.012 – 0.025%, and the chlor content (Cl⁻), of the order of <0,002%. The montmorillonite content (an argillic mineral of extremely high swelling/ shrinking capabilities) is of the order of <2%.

Table 7: Chemical characteristics

pH	SO ₄ %	Cl ⁻ %	Montmorillonite, %	CaCO ₃ %	Redox (mv)
7.8 – 8.2	0.012 – 0.025	<0,002	<2	22 – 30	165 - 225

➤ Unit B – Lavas

The lavas are of doleritic, basaltic and andesitic composition, weakly metamorphosed to zeolitic and lower green schist facies, aphyric and porphyritic, either in the form of pillows or flow sills, mostly intruded by feeder dykes. The latter represents up to 50% of the rock, and consists of multiple doleritic and andesitic dykes, a few cm to almost 2 meters wide, either aphyric or clinopyroxene and/or plagioclase-phyric intrusive rocks.

The lavas, in general, can be characterized as microcrystalline, fine-grained, aphyric to phenocrystalline extrusive, igneous rocks with main petrological type the dolerites, basalts and andesites. The main minerals of which they are composed are quartz and various percentages of amorphous non-crystalline silicon dioxide (glass), plagioclase, diopside, and various types of zeolites. Their upper part is influenced by the atmospheric conditions resulting to their partial weathering, decomposition and disintegration. These phenomena gradually weaken with depth. The highly weathered and decomposed part of lava is usually located to depths of the order of 3-5 meters. Below this depth the healthier rock appears with a much lower degree of weathering and decay and extends beyond depths of 20-30 meters, where the fresh rock is expected. The coloring of the rock is a result of the degree of weathering, decomposition and disintegration and is a good indication of that degree.

The Liquid Limits of the extremely weathered and partially decomposed part of these rocks are of the order of 0 – 32% with corresponding Plasticity Index of the order of 0 – 11%. The Linear Shrinkage is of the order of 0 – 3%. Based on Casagrande Plasticity Chart the decomposed marl can be characterized as soil of intermediate plasticity, soil type ML.

The **natural moisture content** is ranging between 5 and 9% and the **specific gravity** of the order of 2,465 – 2,584 g/cm³.

Table 8: Index Properties – Unit B

Lavas	LL %	PL %	PI %	LS %	Soil type	Moisture Content %
No of tests	2					
Mimumum	0	0	0	0	ML	5 – 9
Maximum	32	29	3	5		
Average	16,0	14,5	1,5	2,5		
Standard Deviation	22,6	20,5	2,1	3,5		
COV%	141	141	141	141		

Density and Strength Parameters

The **Bulk Densities** are of the order of 2,101 – 2,252 g/cm³ with corresponding **Unit Weight** of the order of 20,6 – 22,1 kN/m³. The **Unconfined Compressive Strength (UCS)** is variable and depends on the degree of weathering, decomposition, disintegration and fracturing. The results for the highly weathered lavas are within the range of 0.5 to

2.5 MPa. For less weathered lavas they are of the range of 8 – 25 MPa (from various investigations using core drilling).

From Direct Shear Strength tests on 2 cored samples (taken from lava outcrops) the effective angle of internal friction was established to be of the order of 31 – 35° with effective c of the order of 105 – 220 kPa

Table 9: Strength and density Properties

Lavass	UCS q _u kPa Weathered	c = q _u /2 kPa Weathered	Shearbox		Bulk Density kg/m ³	Unit Weight kN/m ³	Specific Gravity kg/m ³
			Φ' Degrees	c' kPa			
No of tests	4	4	2	2	9	9	2
Minimum	500	250	105	31	2,101	20,6	2,465
Maximum	25000	12500	220	35	2,252	22,1	2,584
Average	9000,0	4500,0	162,5	33,0	2,202	21,6	2,525
Standard Deviation	11128,0	5564,0	81,3	2,8	0,05	0,5	0,1
COV%	124	124	50	9	2	2	3

Based on BS5930 the lavas at this site can be classified as extremely to very weak rock down to the investigated depth, as explained below.

Table 10: Rock classification based on Unconfined Compressive Strength (UCS) in accordance with CYS EN ISO 14689-1:2003

Term for use in field or based on measurement	Definition for field use	Definition on basis of Unconfined Compressive Strength MPa
Extremely weak	Can be indented by thumbnail. Gravel sized lumps crush between finger and thumb.	0.6 – 1.0
Very weak	Crumbles under firm blows with point of geological hammer. Can be peeled by a pocket knife	1 – 5
Weak	Can be peeled by a pocket knife with difficulty. Shallow indentations made by firm blow with the point of geological hammer.	5 – 25
Medium strong	Cannot be scraped with pocket knife. Can be fractured with a single firm blow of geological hammer.	25 – 50
Strong	Requires more than one blow of geological hammer to fracture.	50 – 100
Very strong	Requires many blows of geological hammer to fracture.	100 – 250
Extremely Strong	Can only be chipped with geological hammer.	>250

Based on BS EN ISO 146891:2003 4.2.7, Table 5.

Elastic Parameters – Ground Coefficients

Based on in situ and laboratory tests results, as well as on experiences, the following parameters could be proposed.

Table 11: Ground Coefficients

Poisson's ratio, ν	K _s kN/m ³	E _s kN/m ²
0,25	>100.000	>100.000

Chemical characteristics

The pH is of the order of 7.7 – 7.9. The sulphates content (SO_4), are of the order of <0,02%, and the chlor content (Cl^-), of the order of <0,001%. The montmorillonite content (an argillic mineral of extremely high swelling/ shrinking capabilities) is of the order of <2% for the extremely weathered lavas.

Table 12: Chemical characteristics

pH	SO_4 %	Cl^- %	Montmorillonite, %	CaCO_3 %	Redox (mv)
7.7 – 7.9	<0.02	<0,001	<2	20 – 30	180 - 220

5.4 Electrical Resistivity and Corrosivity

Electrical Resistivity

Laboratory electrical resistivity survey was performed using the equipment which is presented in Appendix 1. The location of the sampled Points are shown in Figure No. 11 and 12 and the results on table No. 7 below. The survey was carried out in such a manner that good site coverage is achieved. The resistivity profile down to 1.50 meters could be obtained, as shown on the table below.

Table 13: Electrical Resistivity Results

TP	Depth m	Soil Description	Moisture content (%)	Temperature Water ($^{\circ}\text{C}$)	Electrical Resistivity (Ωm)
1	0,00-0,70	Soil Cover	8.3	26.0	117.5
1	0,70-2,10	Soil Cover + Weathered Lava fragments	8.5	25.5	1109.7
3	0,00-1,00	Soil Cover	8.1	25.8	593.0
3	1,00-1,60	Soil Cover + Weathered Lava fragments	8.2	25.8	1143.6
6	0,00-1,20	Soil Cover	8.0	26.0	377.5
6	1,20-1,50	Soil Cover + Weathered Lava fragments	8.1	25.8	1117.0

The above results show that the soil cover exhibits lower ER than the lavas.

Soil corrosivity

Soil corrosivity depends on several factors, like soil type, resistivity, pH, redox potential, moisture content, presence of groundwater, homo/inhomogeneity, soil texture, aeration sulphate/sulphides content, and presence of bacteria. Based on the first four factors the following was established and can be used for the decision on metal finish. The results, using several methods, are presented on the tables below.

Table 14: Soil corrosivity based on Electrical Resistivity

Soil type	Soil Resistivity Ωm	Corrosivity	
		NACE	ASTM
Lava and superficial soil	>117	Negligible	Very mildly corrosive

Table 15: Soil corrosivity rating based on pH

Soil type	pH	Corrosivity
Lava and superficial soil	7.5 – 12.0	Moderate

Table 16: Soil corrosivity using Resistivity and pH in accordance with EN 12501-2:2003

Soil type	Soil Resistivity Ωm	pH	Corrosivity
Lava and superficial soil	>117	6.0 – 9.5	Low

Table 17: Soil corrosivity using Redox Potential, mV in accordance with Starkey and Wight.

Soil type	Soil Redox Potential	Corrosivity
Lava and superficial soil	100 – 200	moderate
	200 – 400	Slight

Most of the results are >17 and therefore are considered slightly corrosive.

5.5 Thermal Conductivity

The results of the Thermal Conductivity/Resistivity, carried out in the laboratory using the instruments as shown in Appendix 1, on the same samples used for the electrical resistivity.

Table 18: Thermal Conductivity/ Resistivity Results

TP No.	Depth (m)	Description	Moisture content (%)	Wet Density (gr/cm ³)	Dry Density (gr/cm ³)	Thermal Conductivity (W/mK)	Thermal Resistivity (mK/W)
1	0,00-0,70	Soil Cover	8.3	1.805	1.667	0.205	2.785
1	0,70-2,10	Soil Cover + Weathered Lava fragments	8.5	2.144	1.976	0.235	4.555
3	0,00-1,00	Soil Cover	8.1	1.729	1.599	0.625	1.610
3	1,00-1,80	Soil Cover + Weathered Lava fragments	8.2	2.216	2.048	0.739	2.824
6	1.00-1.20	Soil Cover	8.0	1.670	1.546	0.450	2.336
6	1,20-1,50	Soil Cover + Weathered Lava fragments	8.1	2.101	1.944	0.519	4.680

Soil Chemistry

Three samples were analyzed for pH, Redox, Acid soluble SO₄ and Acid soluble Cl⁻ with the following results.

Table 19: Chemical Results

TP No.	Depth (m)	Description	pH (1:5)	Redox (mv)	Acid soluble (SO ₄) (% w/w) dry	Acid soluble (Cl ⁻) (% w/w) dry
1	0,00-0,70	Soil Cover	7.8	225	<0.02	<0.0018
3	0,00-1,00	Soil Cover	7.7	187	<0.02	<0.0018
6	1.00-1.20	Soil Cover	7.9	220	<0.02	<0.0018

6 FOUNDATION CONDITIONS AT THE STRUCTURES SITES - GEOTECHNICAL CONSIDERATIONS FOR FOUNDATIONS.

6.1 Foundation Conditions

Foundations should be designed to support structural loads with an adequate factor of safety with respect to the soil parameters at each site, so that settlements remain within tolerable limits. Taking into consideration the type of structures to be erected, the site offers good foundation conditions with allowable bearing capacities higher than the loads to be imposed on the ground, at least 180 kPa in the Superficial soil, and more than 350 kPa on the lava bedrock.

6.2 Types of foundations to be studied to this project

Foundations for structures like the ones under study, 3 foundation types are in use:

- Driven piles,
- Cast in situ piles and
- slabs.

Driven piles cannot be used where the lava bedrock is close to or outcrops on the surface and will buckle the piles. Cast in situ piles and slabs, either prefabricated or cast in situ, should be used in most of the area, although in part of it, about 30-40%, driven piles would also be used.

In the area with lava close to the existing ground surface or right on the surface, predrilling might be applied to insert the piles. In the area of soil cover >one-meter thick, driven piles could be used wherever possible, as shown on the relevant figure, and in the rest of the area the other two types could be used.

Whatever the type of foundation, it should be placed to such a depth, so that the structures are protected from overturning due to lateral wind loads. The final foundation depth should

be decided by the engineers of the project taking into consideration the load of the structures as well as the external loads and moments acting on the structures and the foundations. The driven piles might go down to at least 1.2 meters, but the rest might be shallower since they will be placed within or on top of the bedrock.

Since the loads of the structures are very small, no notable total or differential settlements are expected (<5mm). Furthermore, in most of the area, the settlements will be immediate due to the granular nature of the ground. What is prudent is to apply an effective drainage system so that storm water is not eroding the ground surface and is not allowed to infiltrate into the ground that may provoke settlements.

In summary, the following facts should be considered during the foundation design:

- The density and particle size distribution variability of the soils.
- The velocity of the wind should also be considered for the design of the foundations since it may cause uplift of the shallow foundations.
- Apply effective measures to avoid soil surface erosion and infiltration.

6.3 Bearing Capacity and Foundation Conditions

The choice of the allowable bearing pressure is not only a matter of computing but mainly of the correct assessment of the overall geological/geotechnical conditions prevailing at the site. The exercise of a general site judgment by the foundation engineer is, therefore, a must and critical conditions like plasticity characteristics, weathering, density variability, hydrogeological conditions and geological history of the site should be taken seriously into consideration.

Allowable Bearing Pressures for Granular Soils

Down to the lava bedrock all soil types could be considered as granular materials. It should be noted that the bearing pressures of granular materials are quite high in terms of bearing capacities. However, the choice of the allowable bearing pressures for foundation on granular soils is controlled by settlement rather than by strength. The determination of bearing capacity was based on the empirical methods of TERZACHI-PECK (1948), MEYERHOF (1965, 1968) and PECK, HANSON, THORNBURN (1976), which were slightly modified by later scientists.

All the methods above are based on the relation between the soil pressure, which produces a given settlement and the N values from the SPT. To avoid the problems of the density variability, the Allowable Bearing Pressure, which can produce 25mm settlement, is determined for the footing to be subjected to the greatest design load using the lowest value of SPT.

For this study the allowable bearing pressures of the soils at the site, using the DCP converted to SPT results, were determined with a factor of safety of 3. The DCP results are variable and depend on coarseness/fineness of the material. They cannot be reliable. The results of the calculation of the allowable bearing pressures based on DCP/SPT results are presented on table 4 below. Much more tests, than those presented below, were performed but were masked due to the presence of coarse material.

Allowable Bearing Pressures of the order of 130 – 305 to >600 kPa were calculated using this method, as shown on table below.

Table 20: Allowable Bearing Pressures

TP No / Depth m	N (measured)	Density	A.B.P (kPa)
1 / 0,5-0.95 m	13	Medium dense	140
2 / 0.6-1.05 m	12	"	130
3 / 1.0-1.45 m	16	"	170
4 / 1.0-1.45 m	20		215
5 / 1.0-1.45 m	22	"	240
6 / 1.0-1.20 m	Refused >60	Dense	>600
7 / 1.0-1.20 m	Refused >60	"	>600
8 / 0.5	Refused >60	"	240
9 / 0,5-0.95 m	28	Medium dense	305
10 / 0.6-1.05 m	25	"	235

For very light constructions like the one under study, the bearing capacity is not a matter of concern, not even the settlements, which are to be very small and immediate anyway. The problem to deal with in this case is the surface erosion of the soil due to storms. So, further to an effective compaction, a slight resurfacing of the sites within the frame of an effective drainage system should be undertaken so that the runoff of stormwater is well drained away. Infiltration into the ground should be minimized and conditions to allow free and easy drainage should be designed by specialists.

Table 21. Proposed Allowable Bearing Pressures at foundation level

Group	Range of ABP kPa	Proposed ABP at 0.5-1.2 m kPa
Soil cover	130 – 305	130
In situ Lava	>600	300

Based on the results presented above, it could be concluded that the site offers relatively good foundation conditions for the proposed structures with allowable bearing pressures of at least 130 kPa, quite higher than the loads to be distributed on the ground. The only concern, in this case, is the possible settlements of the soil, but they should be small and immediate. The main task is to protect the soil

- from surface runoff and surface soil erosion, as well as
- from water infiltration into the sub-ground that could provoke settlements.

For road construction the following parameters were established.

In accordance with the USC (Unified Soil Classification) the fine soils at the site can be categorized as GM, GC and in accordance with AASHTO/ ASTM A – 2 – 6 and A – 2 – 7.

The Laboratory results for CBR values, established in accordance with CYS EN 13286-47 was found to be >11% due to the high content of coarse material.

6.4 Design Parameters

Coefficient of Subgrade Reaction and Elastic Modulus

These parameters were established using various methods based on SPT, laboratory results and experiences. They are presented on the following table.

Table 22: Proposed Soil subgrade reaction and deformation parameters

Soil Type	Vertical Subgrade Reaction Ks (MN/m ³)	Es (MN/m ²)	Poisson Ratio
Soil Cover	30*	20	0.30
In situ Lava	>100	>100	0.25

* Indicative: To be calculated by the foundation engineers based on foundation type, loads and dimensions.

Table 23: Angle of internal friction, the design compressive strength and the cohesion established on the basis of RQD and Unconfined Compressive Strength for rocks (After Tomlinson).

RQD	Design Compressive Strength kPa	Cohesion kPa	Angle of Friction
Lava in situ			
0-70	0.33 q Lower value: 165 Max. value: 8250 Aver. Value: 2970	0.1 q Lower value: 50 Max. value: 2500 Aver. Value: 900	30
	Proposed: 165	Proposed: 50	Proposed:30

q: Unconfined Compressive Strength

Friction Angle (from lab and in situ tests)

The friction angle as determined with shearbox, triaxial testing and SPT is shown on the following table.

Table 24. Angle of internal friction

Soil Type	ϕ^0	Overall Proposed ϕ^0
Soil Cover	29 – 34	31
In situ Lava	31 – >40	35

Table 25: Specific Gravity, Bulk Density, Unit Weight

Soil Group	Specific Density g/cm ³	Bulk Density g/cm ³	Unit Weight kN/m ³	UCS kPa	Shear Strength, c kPa
Soil Cover	2.285	1.733	17.0	0	<10
In situ Lava	2.525	2.202	21.6	>500	>105

7 EXCAVATION CONDITIONS

Limited earthworks for site cleaning, and some site levelling, are needed. They can be carried out with conventional excavators up to 50 tons. Predrilling for piles, if chosen, is also an easy operation.

8 HYDROGEOLOGICAL CONDITIONS

No ground water is present at shallow depth. In general, it should be stated that the Lavas have very low primary permeability when fresh, however at the uppermost part a secondary permeability is developed along highly weathered and fragmented part. The soil cover is of quite higher permeabilities. Laboratory falling head permeability tests on soil samples were carried out on three samples with results as follow:

- $9.0 \cdot 10^{-4}$ cm/sec
- $4.2 \cdot 10^{-3}$ cm/sec
- $5.5 \cdot 10^{-3}$ cm/sec
- The permeabilities for the uppermost part of the bedrock may ranges from $3.3 \cdot 10^{-4}$ cm/sec to $2.5 \cdot 10^{-3}$ cm/sec

It is, therefore, crucial to apply a proper rainwater management system to prevent infiltration into the soil and furthermore to prevent erosion of the surface soil as already mentioned above.

Table 26: Permeability and drainage characteristics of soil (Terzaghi et al., 1996)

cm/s	10 ⁰	10 ⁻¹	10 ⁻²	10 ⁻³	10 ⁻⁴	10 ⁻⁵	10 ⁻⁶	10 ⁻⁷	10 ⁻⁸	10 ⁻⁹	10 ⁻¹⁰
				Gravel	Sand	Weathered Lava					
Drainage	Good							Poor		Practically Impervious	
Soil types	Clean Gravel		Clean sands, clean sand and gravel mixtures					Very fine sands, organic and inorganic silts, mixtures of sand silt and clay, glacial till, stratified clay deposits etc		“Impervious” soils, e.g. homogenous clays below zone of weathering.	
						“Impervious” soils modified by effects of vegetation and weathering.					

9 ARCHAEOLOGICAL CONDITIONS

No signs of archaeological monuments or remnants were observed during the walkover of the site.

10 GEOLOGICAL HAZARDS

The main geological hazards that could be examined for the site under study are the following:

- Seismic hazard
- Active faults
- Flood hazard
- Karst hazard
- Liquefaction

Seismic risk

In accordance with the National Annex of Eurocode CYS EN 1998-1: 2004, the area is found within seismic zone 2 with peak ground acceleration 0.20. Based on table 3.1 of the Eurocode, the superficial soils in the investigated area are variable and could be classified as Ground Types D and the bedrock as ground type A.

Table 27: Ground Types in accordance with CYS EN 1998-1

Ground Type	Description of Soil Horizon and depth	Parameters		
		Vs30 (m/s)	NSPT	Cu (kPa)
A	Lava	> 800	>60	>250

Active faults

Small risk due to presence of active faults at distance <5 Km.

Flood risk

No such risk exists. However, an effective drainage system should be designed so that infiltration of storm water into the ground and erosion are avoided.

Karst risk

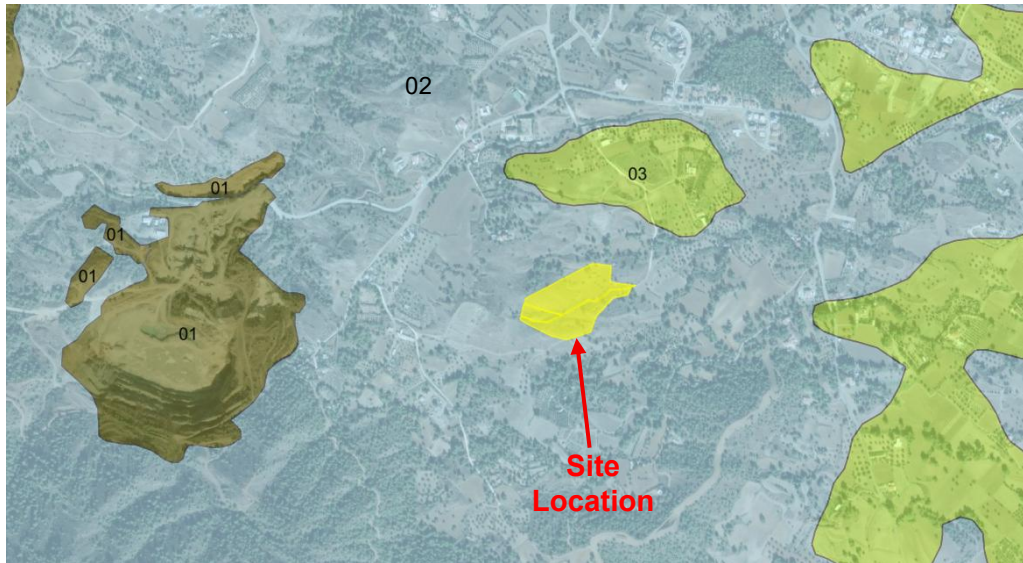
No such risk exists.

Liquefaction

No such risk exists.

Geological Suitability

In accordance with GSD's Geological Suitability maps the area is classified as Zone 2, see figure 8 below.



ZONE 01

The area is threatened by geohazards and identified as an area with instability and risk. These geohazards have very negative and destructive results on the built environment. In this zone, it is required that a geological/geotechnical investigation is executed for all development/ buildings/ structures/ additions/ plot development/ wind farms (aeolian parks)/ photovoltaic parks. Small scale agricultural infrastructure like fences, anti-hail nets, storage facilities (up to 30 m²), water/waste water reservoirs and greenhouses are exempt. In **Zone 01A** special requirements are applied by the licensing authority.



ZONE 02

In this zone, it is required that a geological/geotechnical investigation is executed for all construction projects, except for plot development, photovoltaic parks and structures consisting of less than 2 floors, which do not include basements or swimming pools (the number of floors include the ground floor or open ground floor (pilotis)). More specifically, a geological/geotechnical investigation is required for:

- New building/structure consisting of more than 2 floors, or/and including basement/basements, independent of the total number of floors (the ground floor and open ground floor (pilotis) are considered in the total number of floors),
- Construction of a basement or basements without other ground level, open ground level or multi-level structure,
- Addition of a floor or floors in an existing building/structure, in the case that the total number of floors will result in more than two (the ground floor and open ground floor (pilotis) are considered in the total number of floors), or any other addition which will increase the weight of the structure. In this case, the licensing authority may request the opinion of the Cyprus Geological Survey,
- Construction of a swimming pool in a new or existing site,
- Wind Farms / Aeolian Parks.



ZONE 03

There is no indication for any geohazards threatening the built environment. A geological/geotechnical investigation for geohazards is not required. A geological/geotechnical investigation though, provides the engineer with data about the geological and geotechnical conditions of the area of the project for the successful design of the excavation, foundation and stability works.

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Figure 8. Geological suitability

11 CONCLUSIONS

General Geology: The area is found within “The Eruptive Sequence of the Troodos Ophiolite Complex” represented by The Lower Pillow Lavas.

Due to the high degree of weathering, decomposition and disintegration, residual soils have been developed and alluvial/colluvial material was accumulated on top of the above formations and along the slopes and streams.

Part of the area to be developed is covered with soil the thickness of which ranges from a few centimeters to a couple of meters. The rest is occupied by lava outcrops, which are highly weathered and partially decomposed and disintegrated.

The morphology is mild with gentle slopes and small streams.

Site geology and geotechnical conditions at the solar park site: From the geological point of view the materials encountered in the whole of the investigated site are represented by Lower Pillow Lavas, which are in part of the area covered with superficial, recent soil, mostly residual and colluvial. From the geotechnical engineering point of view, they can be grouped into two Units:

- C. The Soil Cover and
- D. The in-situ Lavas, representing the bedrock of the area.

Geotechnical conditions at the solar park site: About 30 to 40% of the area to be developed is covered with soil, which consists of weathered, variable sized and shaped angular lava fragments embedded in a reddish brown, clayey, silty sandy matrix. The lava fragments are of fine gravel to cobble size, occasionally larger.

Hydrogeology: No groundwater is envisaged at shallow depth. The permeability of the soil cover is high and that of the bedrock medium on top and then low.

The electrical resistivity is high (mostly $>117 \Omega m$) due to the rocky character of the both the soil and the Lava. It varies depending on grains size, density, weathering, and moisture content. The results show that the more the gravel fraction and the lower the weathering of lavas the higher the electrical resistivity.

The **soil corrosivity** is mostly negligible/low to occasionally moderate, depending on the method of evaluation.

The Thermal Conductivity is variable, but within the expected limits, depending on various factors, like moisture content, density, mineral and chemical composition of the soils, soil structure and grain size.

Archaeology: No signs of archaeological monuments or remnants were observed during the walkover of the site.

Finally, the following should be pointed out:

- Taking into consideration the type of the structures to be erected, the site offers very good foundation conditions with allowable bearing capacities at least 130 kPa, quite higher than the loads to be imposed on the ground.
- All types of foundation, shallow pad footings or piles (partially driven and partially predrilled), could be applied for this case.
- Whatever the choice of the type of foundation is, it should be placed to such a depth (≥ 1.0 m in the case of piles and ≥ 0.5 m in the case of pad foundation), so that they are protected from overturning due to lateral wind loads. The final foundation depth and dimensions should be decided by the engineers of the project taking into consideration the load of the structures as well as the external loads and moments acting on the structures and the foundations.
- The roads could be constructed considering CBR of the order of $>11\%$.
- If the proper machinery is used, no issues with the excavations and site cleaning and levelling, if necessary, are envisaged. The excavation conditions, from the geotechnical point of view are classified as “soft/easy” for soil, but medium hard for lavas. Even the small, backhoe excavators, 8-10 tons (~100 HP) could be suitable for small scale excavations or leveling of the site. But for mass excavations excavators of at least 45 tons might be necessary.
- A summary of all geotechnical parameters that can be used for the foundation design are presented on table 28.
- Since the loads of the structures are very small, no notable total or differential settlements are expected ($<5\text{mm}$). Furthermore, the settlements will be immediate due to the granular nature of the soil.
- In accordance with the National Annex of Eurocode CYS EN 1998-1: 2004, the area is found within seismic zone 2 with peak ground acceleration 0.2. Based on table 3.1 of the Eurocode, the rocks in the investigated area could be classified as Ground Type A.
- The main geological hazards that could be considered in the area are:
 - The moderate seismic risk
 - No high flood risk is envisaged, but **an effective drainage system should be designed so that surface erosion and infiltration of storm water into the ground are avoided, preventing thus settlement**, because infiltration of storm water into the ground and the alternating wet and dry conditions may provoke loosening and settlement of the soils with negative effects on the structures’ foundation. An effective storm water management, therefore, could minimize this risk.

Table 28: Geotechnical Characteristics of the soils

PARAMETERS	Unit A Soil Cover	Unit B Lavas
SPT	9 – 25	> 60
Φ	29 – 34 ⁰ Proposed: 31 ⁰	31 – >40 ⁰ Proposed: 35 ⁰
LIQUID LIMITS	20 – 36 %	0 – 32 %
PLASTICITY INDEX	8 – 18 %	0 – 3 %
LINEAR SHRINKAGE	5 – 10 %	0 – 5 %
ALLOWABLE BEARING PRESSURE	130 kPa at 0.5 – 1.2m	300 kPa at 0.5 – 1.2m
MODULUS OF SUBGRADE REACTION Ks	Proposed: 30 MN/m ³	>100 MN/m ³
MODULUS OF ELASTICITY Es	Proposed: 20 MN/m ²	>100 MN/m ²
Unconfined Compressive Strength	54 – 150 kPa (From DCP/SPT)	500 – 25000 kPa
COHESION C _u From UCS (0,5)	27 – 75 kPa	250 – 12500 kPa
COHESION C _u From Shear Box	3 – 10 kPa	105 – 220 kPa
POISON RATIO (μ)	0.30	0.25
pH	7.8 – 8.2	7.7 – 7.9
SO ₄ (%)	0.012 – 0.025	< 0.02
Cl ⁻ (%)	<0.002	<0.001
CaCO ₃ (%)	22 – 30	20 – 30
Redox (%)	165 – 225	180 – 220
Moisture Content (%)	4 – 10	5 – 9
Bulk Density (g/cm ³)	1.550 – 1.915	2.101 – 2.252
Unit Weight (kN/m ³)	15.2 – 18.8	20.6 – 22.1
Specific Gravity (g/cm ³)	2.215 – 2.355	2.465 – 2.584



Figure 9. Seismic Zones of Cyprus

Table 3.1: Ground Types

Ground Type	Description of stratigraphic profile	Parameters		
		v_{s30} (m/s)	N_{SPT} (Blows/30cm)	Cu (kPa)
A	Rock or other rock-like geological formation, including at most 5m of weaker material at the surface.	>800	-	-
B	Deposits of very dense sand, gravel, or very stiff clay at least several tens of metres in thickness, characterized by a gradual increase of mechanical properties with depth.	360-800	>50	>250
C	Deep deposits of dense or medium-dense sand, gravel or stiff clay with thickness from several tens to many hundreds of metres.	180-360	15-50	70-250
D	Deposits of loose-to-medium cohesionless soil (with or without some soft cohesive layers), or of predominantly soft-to-firm cohesive soil.	<180	<15	<70
E	A soil profile consisting of a surface alluvium layers with v_s values of type C or D and thickness varying between about 5m and 20m, underlain by stiffer material with $v_s > 800$ m/s.			
S ₁	Deposits consisting, or containing a layer at least 10m thick, of soft clays/silts with a high plasticity index ($PI > 40$) and high water content.	<100 (indicative)	-	10-20
S ₂	Deposits of liquefiable soils, of sensitive clays, or any other soil profile not included in types A-E or S ₁ .			

National Annex to CYS EN 1998-1:2004 Eurocode 8: Design of Structures for Earthquake Resistance
Part1: General Rules, Seismic Actions and Rules for Buildings

Table 3.2 (CYS): Values of the parameters describing the Type 1 elastic response spectrum

Ground Type	S	T _B (S)	T _C (S)	T _D (S)
A	1.0	0.15	0.4	2.0
B	1.2	0.15	0.5	2.0
C	1.15	0.20	0.6	2.0
D	1.35	0.20	0.8	2.0
E	1.4	0.15	0.5	2.0

APPENDIX 1

1.a EQUIPMENT USED FOR RESISTIVITY SURVEY

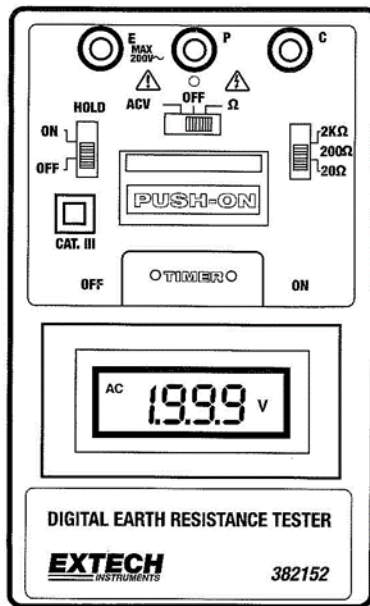
1.b THERMAL RESISTIVITY

User's Manual



Earth Ground Resistance Tester

Model 382152



Introduction

Congratulations on your purchase of the Extech Earth Ground Resistance Tester. This device can measure resistance (in 3 ranges) and AC voltage to 200V. 2mA constant current test signal permits earth resistance tests without tripping current breakers in the circuit under test. Single, momentary tests or automatic 3-minute tests can be selected. This device was designed to meet IEC-1010 (EN 61010) safety standards.

Earth Ground Resistance Testers are useful for:

- Estimating ground resistance of a proposed substation or transmission tower.
- Computing inductive coupling between neighboring power and communication circuits.
- Designing cathode protection systems.
- Geological surveys.

Careful use of this meter will provide years of reliable service.

Warranty

EXTECH INSTRUMENTS CORPORATION warrants this instrument to be free of defects in parts and workmanship for **one year** from date of shipment (a six month limited warranty applies to sensors and cables). If it should become necessary to return the instrument for service during or beyond the warranty period, contact the Customer Service Department at (781) 890-7440 ext. 210 for authorization or visit our website www.extech.com for contact information. A Return Authorization (RA) number must be issued before any product is returned to Extech. The sender is responsible for shipping charges, freight, insurance and proper packaging to prevent damage in transit. This warranty does not apply to defects resulting from action of the user such as misuse, improper wiring, operation outside of specification, improper maintenance or repair, or unauthorized modification. Extech specifically disclaims any implied warranties or merchantability or fitness for a specific purpose and will not be liable for any direct, indirect, incidental or consequential damages. Extech's total liability is limited to repair or replacement of the product. The warranty set forth above is inclusive and no other warranty, whether written or oral, is expressed or implied.

Safety

Please read the following safety information carefully before attempting to operate the meter and use the meter only as specified in this manual.

Environmental safety information

- Do not use meter outdoors when precipitation is likely
- Installation Category III
- Pollution degree 2
- Altitude: 2000 meters max.
- Ambient conditions: 32 to 104°F (0 to 40°C); RH: 80% max
- Observe the following international safety warning symbols

Safety Symbols



Caution: Refer to this manual before using this meter



Dangerous Voltages



Meter is protected throughout by double or reinforced insulation

Organizations that provide rules and guidelines for proper grounding

- The National Electrical Code (NEC)
- Underwriters Laboratories (UL)
- National Fire Protection Association (NFPA)
- American National Standards Institute (ANSI)
- Occupational Safety Health Administration (OSHA)
- Telecommunications Industry Standard (TIA)

Specifications

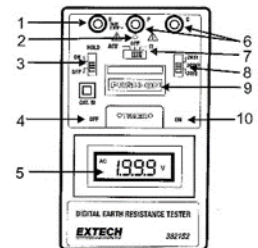
Measurements	Earth ground resistance (in 3 ranges) and AC voltage to 200V
Display	3-1/2 digit (1999 count) LCD with LED test status indicator
Data hold	Freezes latest reading on the LCD display
Automatic Test	3 minute test timer with automatic power down
Measurement triggers	Momentary test or 3 minute test with lock function
Test lead length	Red lead: 50' (15m), Yellow: 33' (10m), Green: 16' (5m)
Power supply	Six 1.5V 'AA' batteries (included)
Low battery indication	LCD displays 'B' icon
Over-range indication	'1' displayed as most significant digit
Safety	IEC-1010 (EN-61010) category III
Weight	27.7oz. (800g) with batteries
Dimensions	6.42 x 3.94 x 2" (163 x 100 x 50 mm)
Supplied accessories	Test leads (3), auxiliary earth bars (2), six 'AA' batteries, and carrying case

Range Specifications

Measurement	Range	Resolution	Accuracy
Earth Ground Resistance 2mA test current (820Hz)	20, 200, 2000Ω	0.01Ω, 0.1Ω, 1Ω	± (2% rdg + 2 d) or ± 0.1Ω whichever is greater
Earth Voltage Frequency: 40 to 500Hz	0 to 200VAC	0.1V	± (1% rdg + 2d)

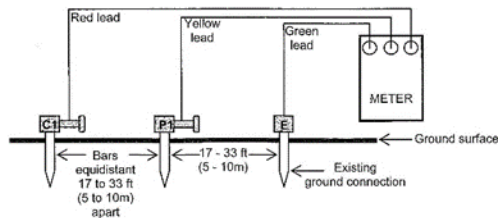
Meter Description

1. Earth ground test lead terminal
2. Test status LED
3. Data Hold switch
4. Automatic 3-min. test OFF button
5. LCD Display
6. C1, P1 Aux bar test lead terminals
7. Function select switch
8. Resistance range select switch
9. Push-button momentary test
10. Automatic 3-min. test ON button



Operation

Test Connection Diagram



Test Preparation and Setup

1. Connect the test leads to the meter as follows:
 - Green lead to the 'E' terminal
 - Yellow lead to the 'P' terminal
 - Red lead to the 'C' terminal
2. Drive the Auxiliary Earth Bars C1 & P1 (included) into the ground. Align the bars equidistant to the existing Earth ground connection and in a straight line as shown in the diagram above.
3. Ensure that the bars are between 17 and 33 ft (5 and 10 meters) apart. If the auxiliary bars are placed in close proximity to the ground stake, measurement inaccuracies will result.
4. Connect the clamp ends of the test leads to the Earth bars and existing ground connection spike as shown above.

Earth Voltage Test

1. Set the meter's function switch to the ACV position.
2. Press the PUSH-ON button to perform a single, momentary test.
3. Press the PUSH-ON and TIMER ON buttons simultaneously to begin a 3-minute test. The 3-minute test automatically shuts down after 3 minutes. Press the Timer OFF button to end an automatic test at any time.
4. The front panel test status LED will light if the test is operating correctly. If the LED does not light check for problems such as open circuits or overload conditions.
5. Note the LCD reading.
6. Confirm that the voltage measurement is less than 10V AC; otherwise accurate Earth resistance measurements cannot be made. If voltage is present (higher than 10V AC), the source of the voltage must be found and corrected before testing can continue.

Earth Resistance Test

1. Set the meter's function switch to the ohms position and set the range switch to the desired resistance range.
2. Press the PUSH-ON button to perform a single, momentary test.
3. Press the PUSH-ON and TIMER ON buttons simultaneously to begin a 3-minute test. 3-minute tests automatically shut down after 3 minutes. Press the Timer OFF button to end an automatic test at any time.
4. Note the displayed reading. The front panel test status LED will light if the test is operating correctly. If the LED does not light, check for problems such as open circuits or overload conditions.
5. If high resistance is detected, note the value and take appropriate steps to correct the ground connection if necessary.
6. Readings "1" ...Ω are typical when the test leads are not connected to the meter.

Data Hold

- Data Hold freezes the latest measurement reading on the LCD.
1. Select ON from the Data Hold selector switch to turn Data Hold ON.
 2. The reading that was on the display when the switch was set is now held on the LCD.
 3. The Data Hold function does not retain measurement data if the meter is turned off.
 4. Select OFF from the Data Hold selector switch to return meter to normal operation.

TLS-100 Portable Meter

User Manual



 **THERMTEST**
PORTABLE



TLS-100 User Manual

IN THE BOX

When first receiving your new TLS-100 Meter, please take a few moments and check that your shipment includes all of the following items:

- TLS-100 Controller
- 100 mm Needle Probe
- Sensor Insert (1)
- Sensor Sleeves (3)
- Drill Bits (2)
- Batteries
- Carrying Case
- Thermal Paste
- MicroSD memory card (inserted in TLS-100)
- Verification Sample and QA Certificate



Figure 1. TLS-100 Thermal Conductivity Meter and carrying case.

APPENDIX 2

GEOLOGICAL DESCRIPTION AND PHOTOS OF TRIAL PITS

	
	
	
	
	
Site View	

GEOLOGICAL LOG OF TRIAL PIT

TP No. 1

Project: PLOT 316
Client: ENERGYINTEL SERVICES LTD
Chainage:
Date started: 09/09/2025

Location: SIA
Ground Level: m. a.m.s.l.
Date completed: 09/09/2025

DEPTH BELOW SURFACE m	LOG	DESCRIPTION OF STRATA	REMARKS (SAMPLING)
		The uppermost part ~ 0,4 m used to be cultivated Loose to medium dense, brown residual soil consisting of a mixture of clay, silt and sand with variable amounts of variably sized and shaped (mostly angular) weathered lava fragments. Possibility of pile driving high. Gradually it turns to highly weathered Lava	
0.50			
1.00			
1.50			
2.00			
2.10			
		END OF T.P. AT 2.10 m	
2.50			
3.00			
3.50			

NOTES: Excavator used:

: Pit stability: Stable

: Groundwater observations: G/W not encountered



TP 1

GEOLOGICAL LOG OF TRIAL PIT

TP No. 2

Project: PLOT 316
Client: ENERGYINTEL SERVICES LTD
Chainage:
Date started: 09/09/2025

Location: SIA
Ground Level: m. a.m.s.l.
Date completed: 09/09/2025

DEPTH BELOW SURFACE m	LOG	DESCRIPTION OF STRATA	REMARKS (SAMPLING)
		The uppermost part ~ 0,4 m used to be cultivated Loose to medium dense, brown residual soil consisting of a mixture of clay, silt and sand with variable amounts of variably sized and shaped (mostly angular) weathered lava fragments. Possibility of pile driving high. Gradually it turns to highly weathered Lava	
0.50			
1.00			
1.50			
2.00			
		END OF T.P. AT 2.0 m	
2.50			
3.00			
3.50			

NOTES: Excavator used:

: **Pit stability:** Stable

: **Groundwater observations:** G/W not encountered



TP 2

GEOLOGICAL LOG OF TRIAL PIT

TP No. 3

Project: PLOT 316
Client: ENERGYINTEL SERVICES LTD
Chainage:
Date started: 09/09/2025

Location: SIA
Ground Level: m. a.m.s.l.
Date completed: 09/09/2025

DEPTH BELOW SURFACE m	LOG	DESCRIPTION OF STRATA	REMARKS (SAMPLING)
		The uppermost part ~ 0,4 m used to be cultivated Loose to medium dense, brown residual soil consisting of a mixture of clay, silt and sand with variable amounts of variably sized and shaped (mostly angular) weathered lava fragments. Possibility of pile driving moderate. Gradually it turns to highly weathered Lava	
0.50			
1.00			
1.50			
1.60			
		END OF T.P. AT 1.60 m	
2.00			
2.50			
3.00			
3.50			

NOTES: Excavator used:

: **Pit stability:** Stable

: **Groundwater observations:** G/W not encountered



TP 3

GEOLOGICAL LOG OF TRIAL PIT

TP No. 4

Project: PLOT 316

Location: SIA

Client: ENERGYINTEL SERVICES LTD

Chainage:

Ground Level: m. a.m.s.l.

Date started: 09/09/2025

Date completed: 09/09/2025

DEPTH BELOW SURFACE m	LOG	DESCRIPTION OF STRATA	REMARKS (SAMPLING)
		The uppermost part ~ 0,4 m used to be cultivated <	

NOTES: Excavator used:

: **Pit stability:** Stable

: **Groundwater observations:** G/W not encountered



TP 4

GEOLOGICAL LOG OF TRIAL PIT

TP No. 5

Project: PLOT 316
Client: ENERGYINTEL SERVICES LTD
Chainage:
Date started: 09/09/2025

Location: SIA
Ground Level: m. a.m.s.l.
Date completed: 09/09/2025

DEPTH BELOW SURFACE m	LOG	DESCRIPTION OF STRATA	REMARKS (SAMPLING)
0.50		Hard, brown, moderately to highly weathered Lower Pillow LAVAS.	
		Pile driving not possible / very low possibility	
1.00			
1.40			
1.50		END OF T.P. AT 1.40 m	
2.00			
2.50			
3.00			
3.50			

NOTES: Excavator used:

: **Pit stability:** Stable

: **Groundwater observations:** G/W not encountered



TP 5

GEOLOGICAL LOG OF TRIAL PIT

TP No. 6

Project: PLOT 316
Client: ENERGYINTEL SERVICES LTD
Chainage:
Date started: 09/09/2025

Location: SIA
Ground Level: m. a.m.s.l.
Date completed: 09/09/2025

DEPTH BELOW SURFACE m	LOG	DESCRIPTION OF STRATA	REMARKS (SAMPLING)
0.50		Hard, brown, moderately to highly weathered Lower Pillow LAVAS.	
		Pile driving not possible	
1.00			
1.20			

		Less weathered LAVA	
1.50			
		END OF T.P. AT 1.20 m	
2.00			
2.50			
3.00			
3.50			

NOTES: Excavator used:

: **Pit stability:** Stable

: **Groundwater observations:** G/W not encountered



TP 6

GEOLOGICAL LOG OF TRIAL PIT

TP No. 7

Project: PLOT 316

Location: SIA

Client: ENERGYINTEL SERVICES LTD

Chainage:

Ground Level: m. a.m.s.l.

Date started: 09/09/2025

Date completed: 09/09/2025

DEPTH BELOW SURFACE m	LOG	DESCRIPTION OF STRATA	REMARKS (SAMPLING)
		Hard, brown, moderately to highly weathered Lower Pillow LAVAS. Pile driving not possible / very low possibility	
0.50			
1.00			
1.20			
		END OF T.P. AT 1.20 m	
1.50			
2.00			
2.50			
3.00			
3.50			

NOTES: Excavator used:

: **Pit stability:** Stable

: **Groundwater observations:** G/W not encountered



TP 7

GEOLOGICAL LOG OF TRIAL PIT

TP No. 8

Project: PLOT 316
Client: ENERGYINTEL SERVICES LTD
Chainage:
Date started: 09/09/2025

Location: SIA
Ground Level: m. a.m.s.l.
Date completed: 09/09/2025

DEPTH BELOW SURFACE m	LOG	DESCRIPTION OF STRATA	REMARKS (SAMPLING)
		Hard, brown, moderately to highly weathered Lower Pillow LAVAS. Pile driving down to 0.6 m moderate	
0.50			
0.60			
		Pile driving not possible / very low possibility	
1.00			
1.20			
		END OF T.P. AT 1.20 m	
1.50			
2.00			
2.50			
3.00			
3.50			

NOTES: Excavator used:

: **Pit stability:** Stable

: **Groundwater observations:** G/W not encountered



TP 8

GEOLOGICAL LOG OF TRIAL PIT

TP No. 9

Project: PLOT 316

Location: SIA

Client: ENERGYINTEL SERVICES LTD

Chainage:

Ground Level: m. a.m.s.l.

Date started: 09/09/2025

Date completed: 09/09/2025

DEPTH BELOW SURFACE m	LOG	DESCRIPTION OF STRATA	REMARKS (SAMPLING)
		Hard, brown, moderately to highly weathered Lower Pillow LAVAS. Pile driving not possible	
0.50			
1.00			
		END OF T.P. AT 1.0 m	
1.50			
2.00			
2.50			
3.00			
3.50			

NOTES: Excavator used:

: **Pit stability:** Stable

: **Groundwater observations:** G/W not encountered



TP 9

GEOLOGICAL LOG OF TRIAL PIT

TP No. 10

Project: PLOT 316

Location: SIA

Client: ENERGYINTEL SERVICES LTD

Chainage:

Ground Level: m. a.m.s.l.

Date started: 09/09/2025

Date completed: 09/09/2025

DEPTH BELOW SURFACE m	LOG	DESCRIPTION OF STRATA	REMARKS (SAMPLING)
		Loose to medium dense, brown residual soil consisting of a mixture of clay, silt and sand with variable amounts of variably sized and shaped (mostly angular) weathered lava fragments. Possibility of pile driving high.	
0.50			
1.00			
1.50			
		END OF T.P. AT 1.50 m	
2.00			
2.50			
3.00			
3.50			

NOTES: Excavator used:

: **Pit stability:** Stable

: **Groundwater observations:** G/W not encountered



TP 10

APPENDIX 3

GRAPHICAL PRESENTATION OF LABORATORY TESTS

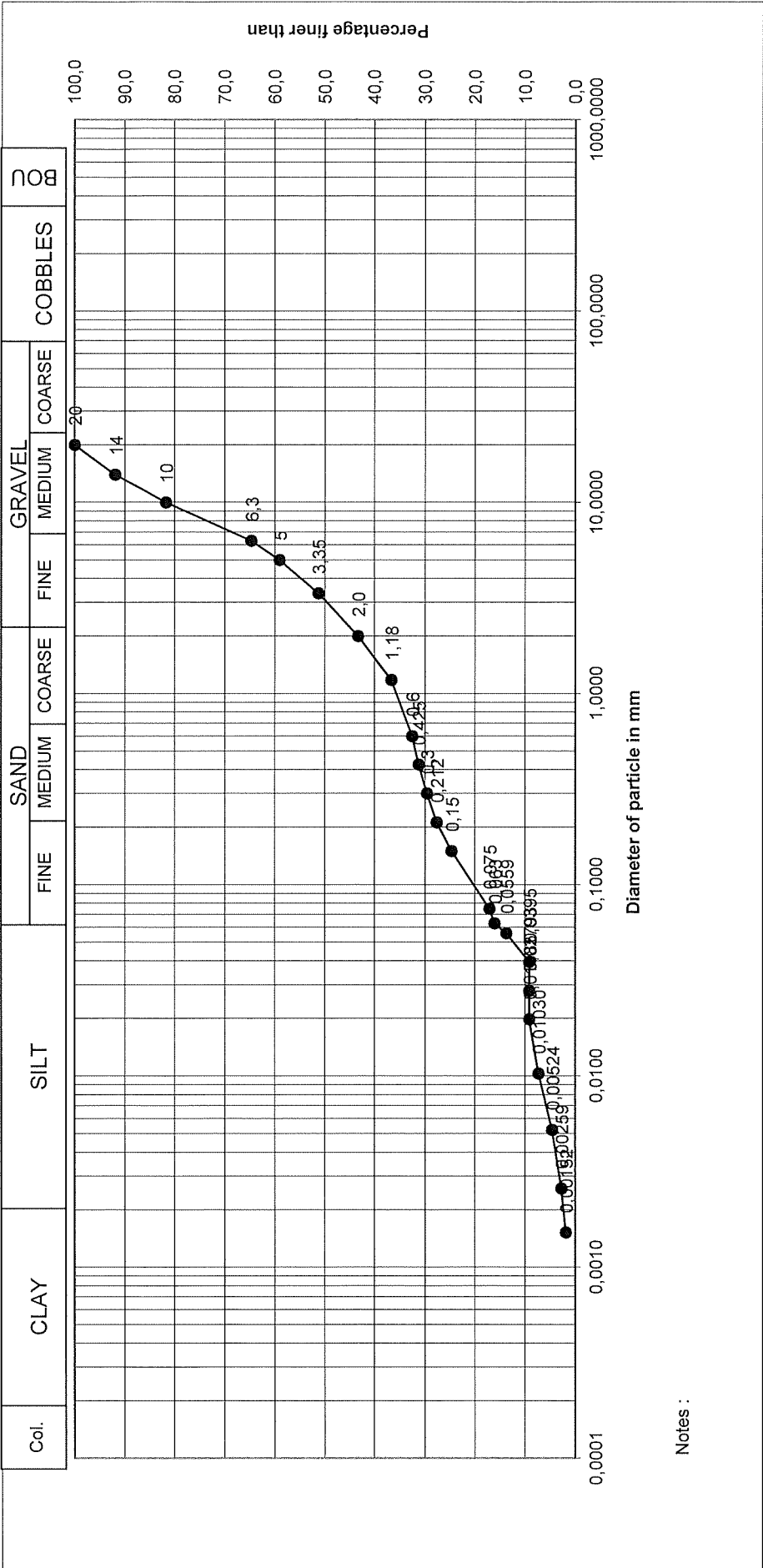
PARTICLE SIZE DISTRIBUTION

CEN ISO/TS 17892-4: 2004

Client:ENERGYINTEL SERVICES LTD
Project: PLOT 316
Site: SHA

Operator : Chr.I.
Date of test completion : 17/09/2025
Description of material : Slightly clayey, silty, very sandy Gravel

TP No. : 1
Depth : 1,00-2,00m



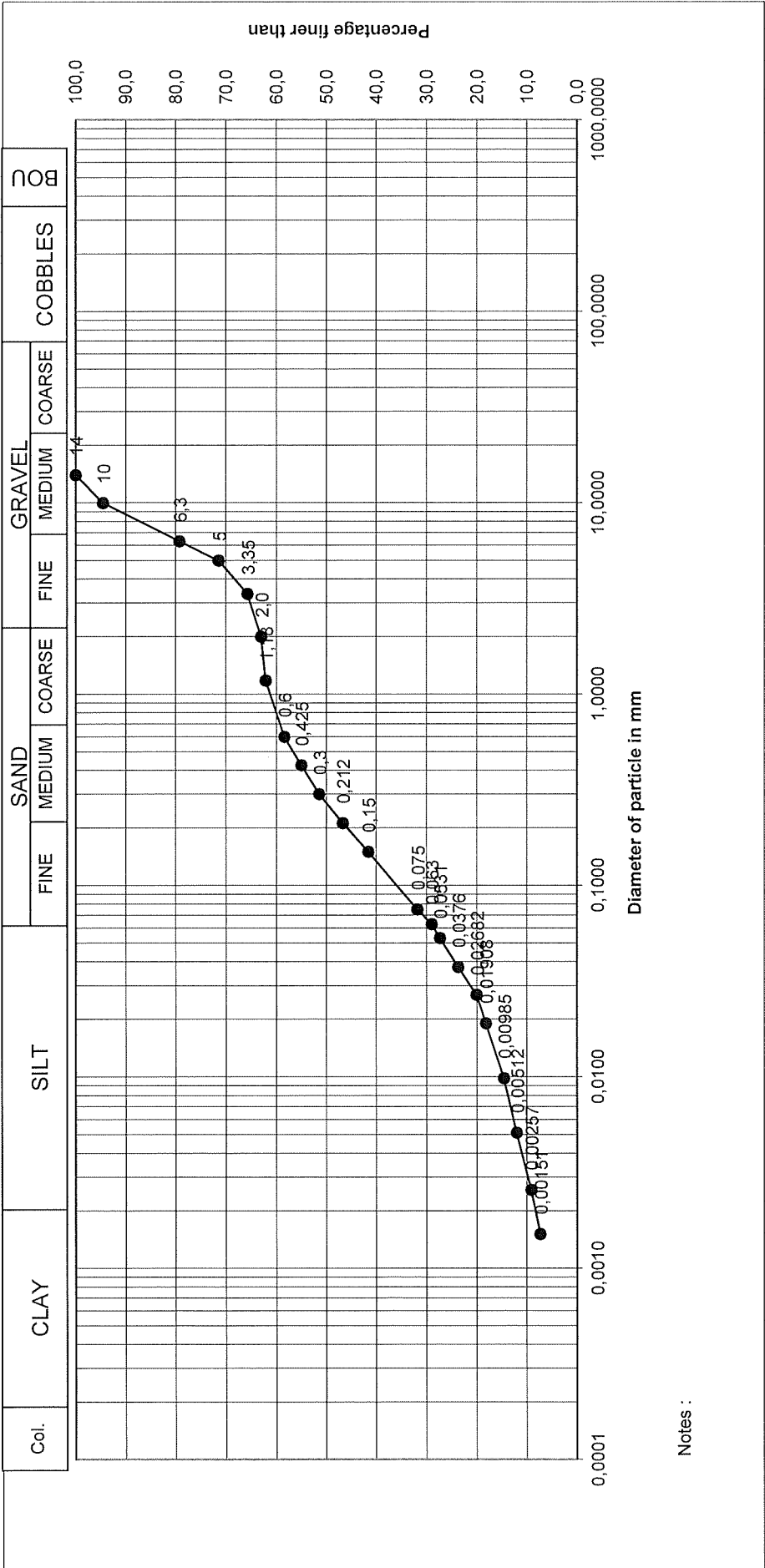
PARTICLE SIZE DISTRIBUTION

CEN ISO/TS 17892-4: 2004

Client:ENERGYINTEL SERVICES LTD
Project: PLOT 316
Site: SHA

Operator : Chr.I.
Date of test completion : 17/09/2025
Description of material : Clayey, very silty, very sandy Gravel

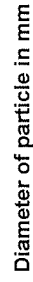
TP No. : 4
Depth : 1,00-1,50m



CEN ISO/TS 17892-4: 2004

Description of material: Slightly clayey, silty, sandy Gravel

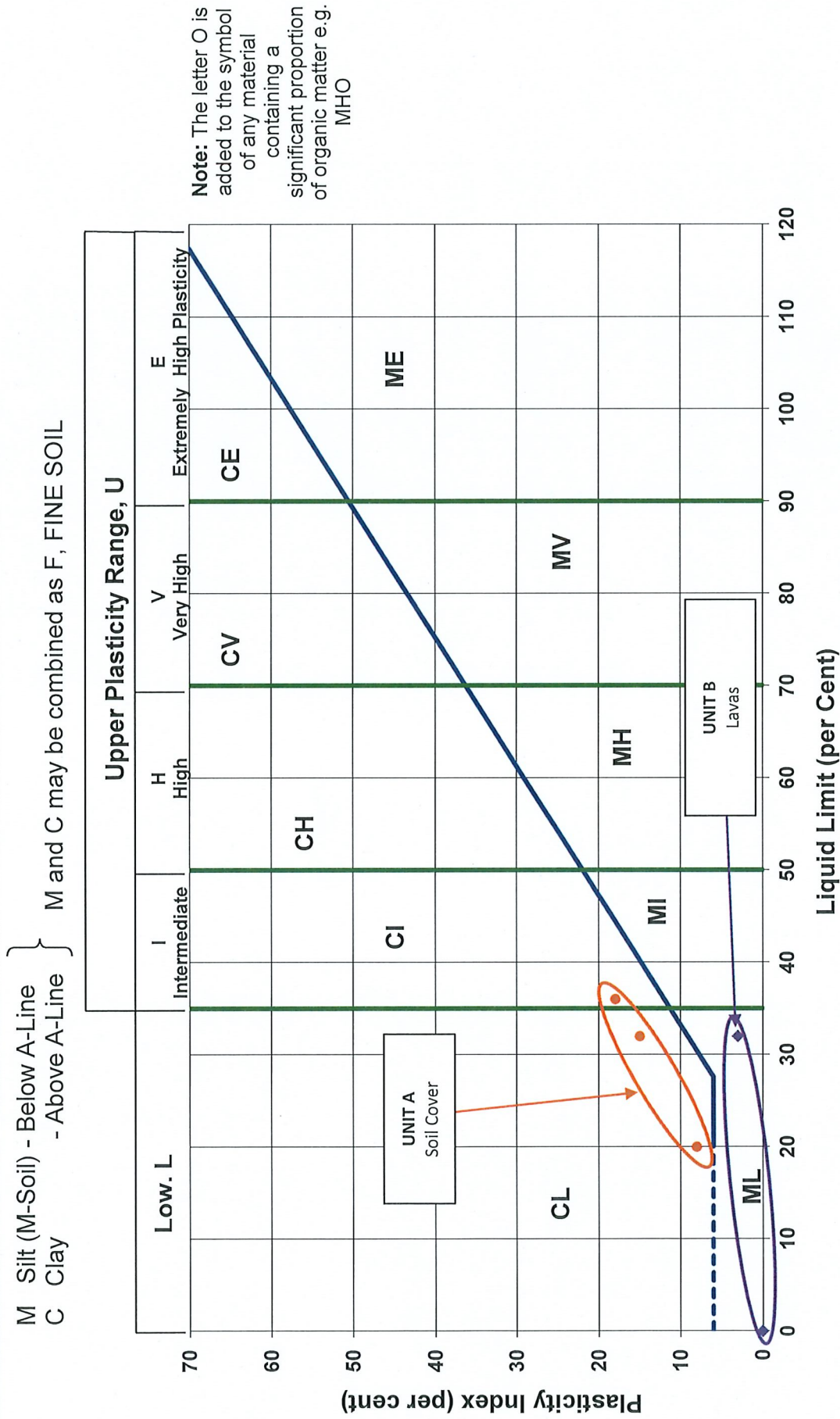
Depth: 0,00-1,00m



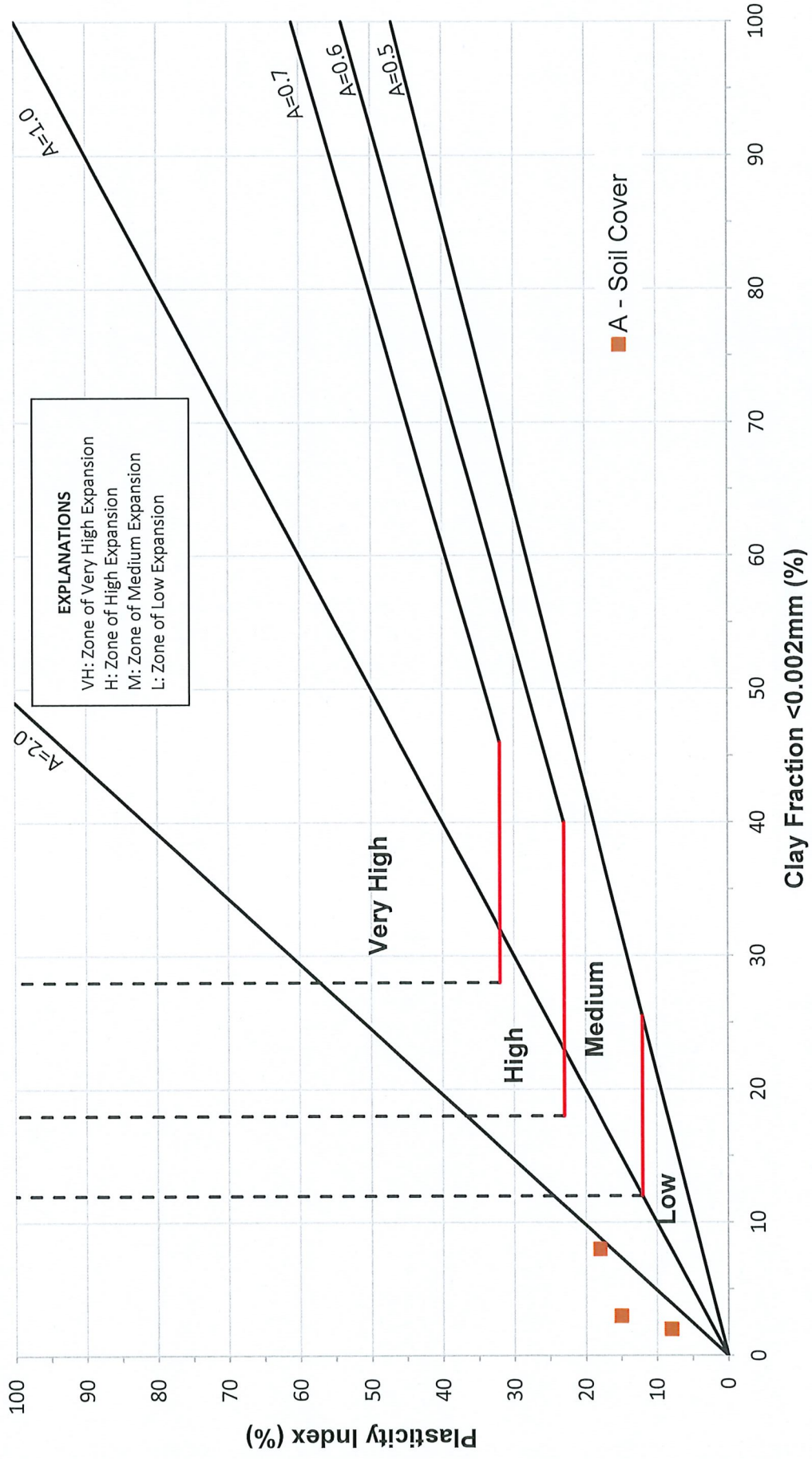
GEOINVEST LTD - ΓΕΩΕΠΕΥΝΑ Tel.: 22-330093, Fax: 22-330118, P.O.BOX: 20476, NICOSIA

Plasticity Chart for the Classification of Fine Soils and Finer Part of Coarse Soils

According to BS 5930:1999



SOUTH AFRICAN EXPANSIVE CLAY CLASSIFICATION CHART ΝΟΤΙΟΑΦΡΙΚΑΝΙΚΟ ΔΙΑΓΡΑΜΜΑ ΚΑΤΑΤΑΞΗΣ (ΔΥΝΗΤΙΚΟΤΗΤΑΣ) ΔΙΟΓΚΩΣΗΣ ΑΡΓΙΛΟΥ



DETERMINATION OF ATTERBERG LIMITS

CEN ISO/TS 17892-12: 2004

Fall Cone Method - 80g/30°

Client:ENERGYINTEL SERVICES LTD

Project: PLOT 316

Site: SHA

Sampler : GEOINVEST L.T.D.

Date of sampling :

Description of material : Slightly clayey, silty, very sandy Gravel

Proportion of sample passing 425 µm BS test sieve, %

Soil equilibrated with water for 24 h.

Trial Pit No : 1

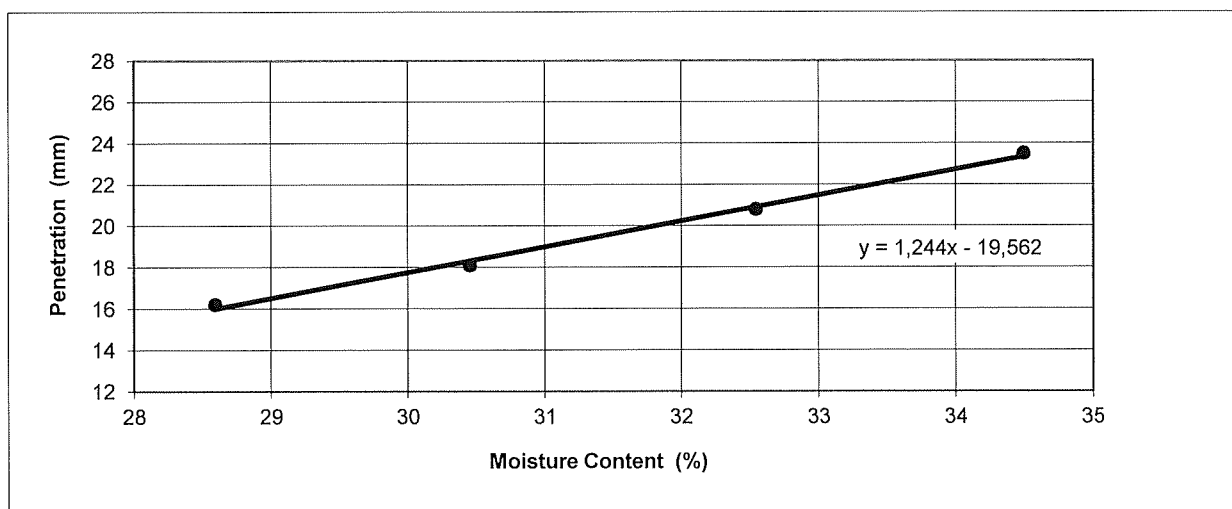
Depth : 1,00-2,00m

Condition of sample : Natural State

Date of test completion : 18/09/2025

Operator : Chr.I.

Test No.	1	2	3	4	5	6
Mass of wet soil + container g	34,58	38,15	42,15	43,65	27,95	27,72
Mass of dried soil + container g	32,63	35,40	38,18	39,49	27,65	27,51
Mass of container g	25,81	26,37	25,98	27,43	25,96	26,17
Mass of moisture g	1,95	2,75	3,97	4,16	0,3	0,21
Mass of dry soil g	6,82	9,03	12,2	12,06	1,69	1,34
Moisture content (w) %	28,59	30,45	32,54	34,49	17,75	15,67
Cone Penetration mm	16,2	18,1	20,8	23,5		
P.L.					16,7	



L.L. = (y+b)/a		P.I. = L.L.-P.L.		CLASSIFICATION	MOISTURE CONTENT
y =	20	L.L. =	32	CL Inorganic clay of low plasticity	
a =	1,244	P.L. =	17		
b =	19,562	P.I. =	15		
LINEAR SHRINKAGE %		7			

GEOINVEST LTD - ΓΕΩΠΕΥΝΑ

Applied Geology Geotechnics-Environmental Engineering

TEL. : 22 - 330093, 330097, 330113 FAX. : 22 - 330118 P.O.BOX : 20476 NICOSIA

DETERMINATION OF ATTERBERG LIMITS

CEN ISO/TS 17892-12: 2004

Fall Cone Method - 80g/30°

Client:ENERGYINTEL SERVICES LTD

Project: PLOT 316

Site: SHA

Sampler : GEOINVEST L.T.D.

Date of sampling :

Description of material : Clayey, very silty, very sandy Gravel

Proportion of sample passing 425 µm BS test sieve, %

Soil equilibrated with water for 24 h.

Trial Pit No : 4

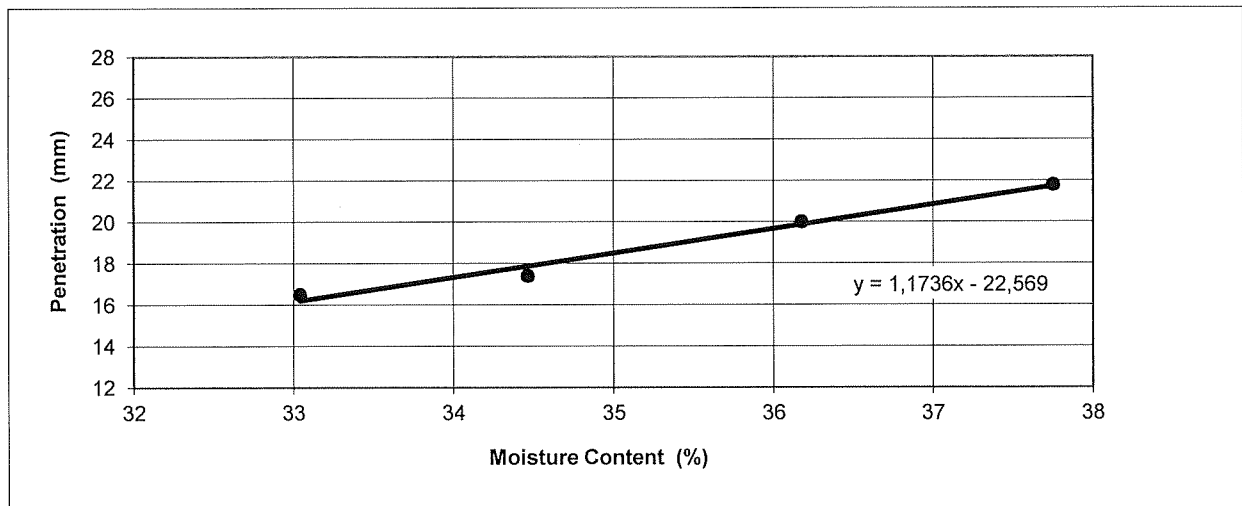
Depth : 1,00-1,50m

Condition of sample : Natural State

Date of test completion : 18/09/2025

Operator : Chr.I.

Test No.	1	2	3	4	5	6
Mass of wet soil + container g	41,03	40,33	39,05	41,52	30,75	30,55
Mass of dried soil + container g	36,88	36,77	35,57	37,19	30,00	29,87
Mass of container g	24,32	26,44	25,95	25,72	25,98	25,89
Mass of moisture g	4,15	3,56	3,48	4,33	0,75	0,68
Mass of dry soil g	12,56	10,33	9,62	11,47	4,02	3,98
Moisture content (w) %	33,04	34,46	36,17	37,75	18,66	17,09
Cone Penetration mm	16,5	17,4	20,0	21,8		
P.L.					17,9	



L.L. = (y+b)/a	P.I. = L.L.-P.L.	CLASSIFICATION	MOISTURE CONTENT
y = 20	L.L. = 36	CI	
a = 1,1736	P.L. = 18	Inorganic clay	
b = 22,569	P.I. = 18	of intermediate plasticity	
LINEAR SHRINKAGE %		10	

GEOINVEST LTD - ΓΕΩΕΠΕΥΝΑ

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DETERMINATION OF ATTERBERG LIMITS

CEN ISO/TS 17892-12: 2004

Fall Cone Method - 80g/30°

Client:ENERGYINTEL SERVICES LTD

Project: PLOT 316

Site: SHA

Sampler : GEOINVEST L.T.D.

Date of sampling :

Description of material : Weathered Lava

Proportion of sample passing 425 µm BS test sieve, %

Soil equilibrated with water for 24 h.

Trial Pit No : 5

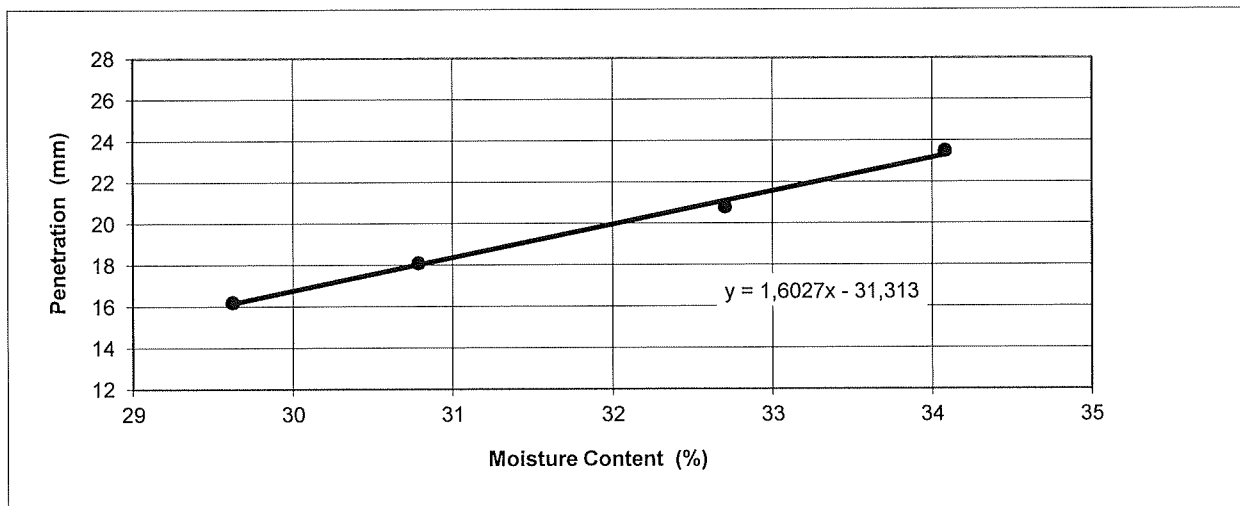
Depth : 0,50-1,50m

Condition of sample : Natural State

Date of test completion : 18/09/2025

Operator : Chr.I.

Test No.	1	2	3	4	5	6
Mass of wet soil + container g	34,65	38,18	42,17	43,60	28,15	27,90
Mass of dried soil + container g	32,63	35,40	38,18	39,49	27,65	27,51
Mass of container g	25,81	26,37	25,98	27,43	25,96	26,17
Mass of moisture g	2,02	2,78	3,99	4,11	0,5	0,39
Mass of dry soil g	6,82	9,03	12,2	12,06	1,69	1,34
Moisture content (w) %	29,62	30,79	32,70	34,08	29,59	29,10
Cone Penetration mm	16,2	18,1	20,8	23,5		
P.L.					29,3	



L.L. = (y+b)/a	P.I. = L.L.-P.L.	CLASSIFICATION	MOISTURE CONTENT
y = 20	L.L. = 32	ML	
a = 1,6027	P.L. = 29	Inorganic silt	
b = 31,313	P.I. = 3	of low compressibility	
LINEAR SHRINKAGE %	5		

GEOINVEST LTD - ΓΕΩΠΕΥΝΑ

Applied Geology Geotechnics-Environmental Engineering

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DETERMINATION OF ATTERBERG LIMITS

CEN ISO/TS 17892-12: 2004

Fall Cone Method - 80g/30°

Client:ENERGYINTEL SERVICES LTD

Project: PLOT 316

Site: SHA

Sampler : GEOINVEST L.T.D.

Date of sampling :

Description of material : Weathered Lava

Proportion of sample passing 425 µm BS test sieve, %

Soil equilibrated with water for 24 h.

Trial Pit No : 9

Depth : 0,50-1,00m

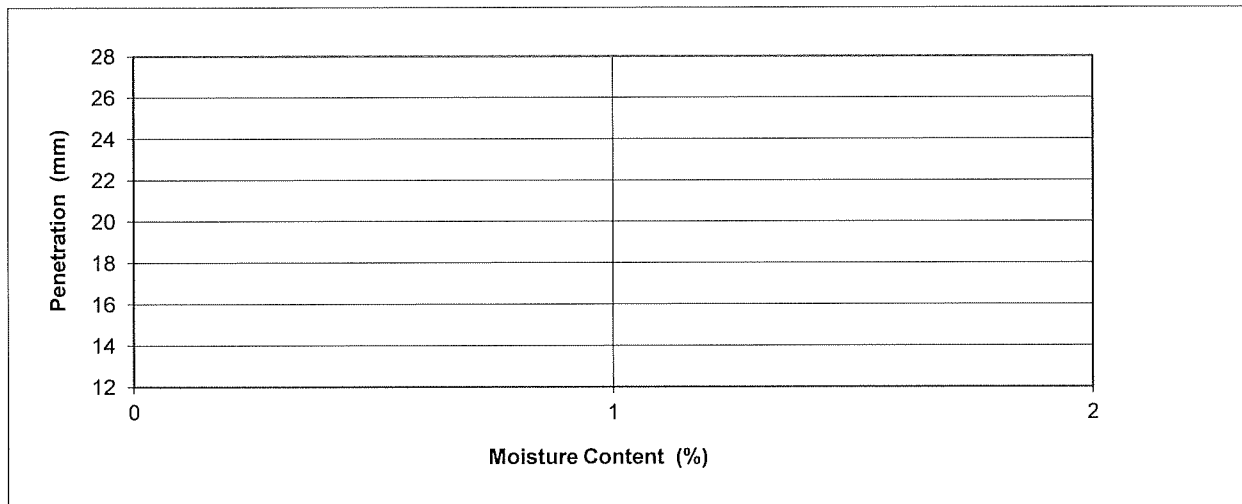
Condition of sample : Natural State

Date of test completion : 18/09/2025

Operator : P.H.

Test No.	1	2	3	4	5	6
Mass of wet soil + container g						
Mass of dried soil + container g						
Mass of container g						
Mass of moisture g						
Mass of dry soil g						
Moisture content (w) %						
Cone Penetration mm						
P.L.						

NOT PLASTIC



L.L. = $(y+b)/a$	P.I. = L.L.-P.L.	CLASSIFICATION	MOISTURE CONTENT
y = 20	L.L. = N.PLASTIC	ML	
a =	P.L. = N.PLASTIC	Inorganic silt	
b =	P.I. = N.PLASTIC	of low compressibility	
LINEAR SHRINKAGE %			

GEOINVEST LTD - ΓΕΩΕΠΕΥΝΑ

Applied Geology Geotechnics-Environmental Engineering

TEL : 22 - 330093, 330097, 330113 FAX : 22 - 330118 P.O.BOX : 20476 NICOSIA

DETERMINATION OF ATTERBERG LIMITS

CEN ISO/TS 17892-12: 2004

Fall Cone Method - 80g/30°

Client:ENERGYINTEL SERVICES LTD

Project: PLOT 316

Site: SHA

Sampler : GEOINVEST L.T.D.

Date of sampling :

Description of material: Slightly clayey, silty, sandy Gravel

Proportion of sample passing 425 µm BS test sieve, %

Soil equilibrated with water for 24 h.

Trial Pit No : 10

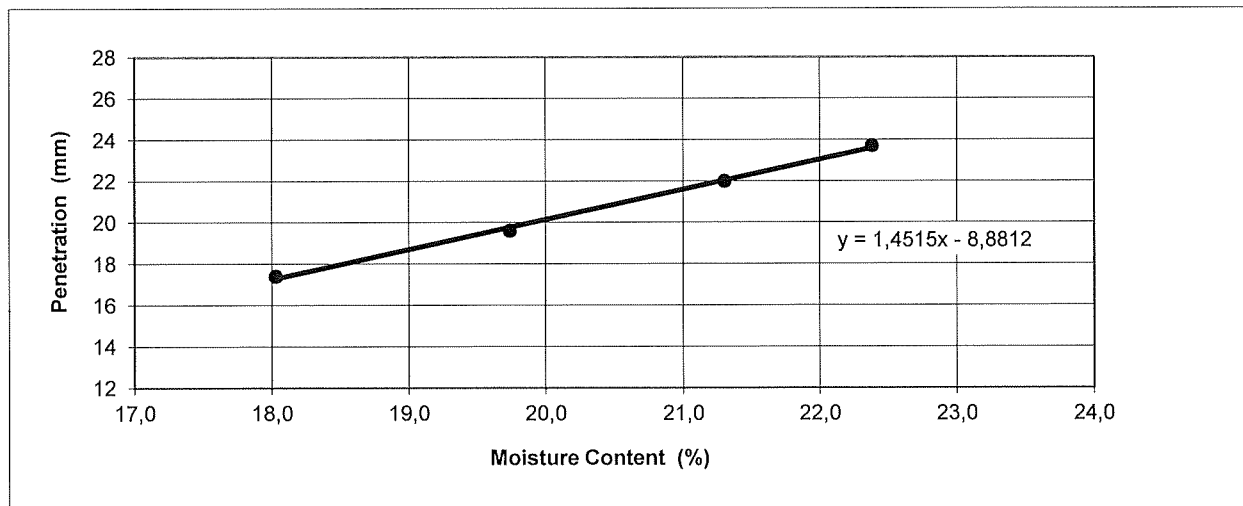
Depth : 0,00-1,00 m

Condition of sample : Natural State

Date of test completion : 18/09/2025

Operator : P. H.

Test No.	1	2	3	4	5	6
Mass of wet soil + container g	52,65	55,01	53,79	58,98	30,35	30,52
Mass of dried soil + container g	48,37	50,19	48,89	52,96	29,80	30,01
Mass of container g	24,63	25,77	25,89	26,06	25,49	25,56
Mass of moisture g	4,28	4,82	4,90	6,02	0,55	0,51
Mass of dry soil g	23,74	24,42	23,00	26,90	4,31	4,45
Moisture content (w) %	18,03	19,74	21,30	22,38	12,76	11,46
Cone Penetration mm	17,4	19,6	22,0	23,7		
P.L.					12,1	



L.L. = (y+b)/a	P.I. = L.L.-P.L.	CLASSIFICATION	MOISTURE CONTENT
y = 20	L.L. = 20	CL	
a = 1,4515	P.L. = 12	Inorganic clay	
b = 8,8812	P.I. = 8	of low plasticity	
LINEAR SHRINKAGE %	5		

GEOINVEST LTD - ΓΕΩΠΕΥΝΑ

Applied Geology Geotechnics-Environmental Engineering

TEL. : 22 - 330093, 330097, 330113 FAX. : 22 - 330118 P.O.BOX : 20476 NICOSIA

Shear Strenght by direct shear (small shearbox) CEN ISO/TS 17892-10: 2004



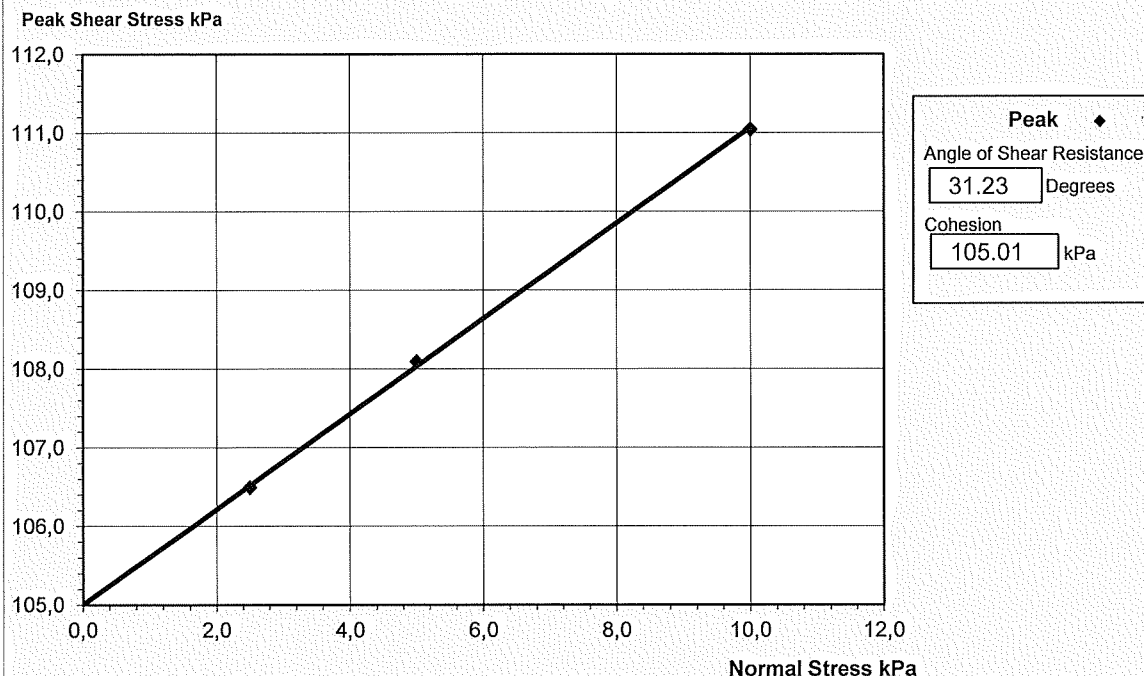
Client:ENERGYINTEL SERVICES LTD
Project: PLOT 316
Site: SHA
Sampler : GEOINVEST LTD

Trial Pit No. : Surface Sample
Depth :
Description of material: Weathered Lava
Date of test completion: 19/09/2025
Operator : Ch.P.

Reference	A	B	C
Applied Normal Stress	2,5 kPa	5,0 kPa	10,0 kPa
Peak Strength	106,5 kPa	108,1 kPa	111,1 kPa
Residual Shear Stress			
Rate(s) of Shear Displacement	Stage 1: 0,9812mm/min	Stage 1: 0,9715mm/min	Stage 1: 0,9521mm/min
Final Height	19,99 mm	19,95 mm	19,94 mm
Bulk Density	2,218 Mg/m ³	2,188 Mg/m ³	2,225 Mg/m ³
Dry Density	2,042 Mg/m ³	2,013 Mg/m ³	2,047 Mg/m ³
Moisture Content	8,6 %	8,7 %	8,7 %
Number of Traverses	1	1	1

Condition of sample: Remoulded
Area : 60 mm x 60 mm

Maximum Shear Stress vs Normal Stress



Shear Strenght by direct shear (small shearbox) CEN ISO/TS 17892-10: 2004



Client:ENERGYINTEL SERVICES LTD

Project: PLOT 316

Site: SHA

Sampler : GEOINVEST LTD

Trial Pit No. :Surface Sample

Depth :

Description of material: Weathered Lava

Date of test completion : 16/09/2025

Operator : Ch.P.

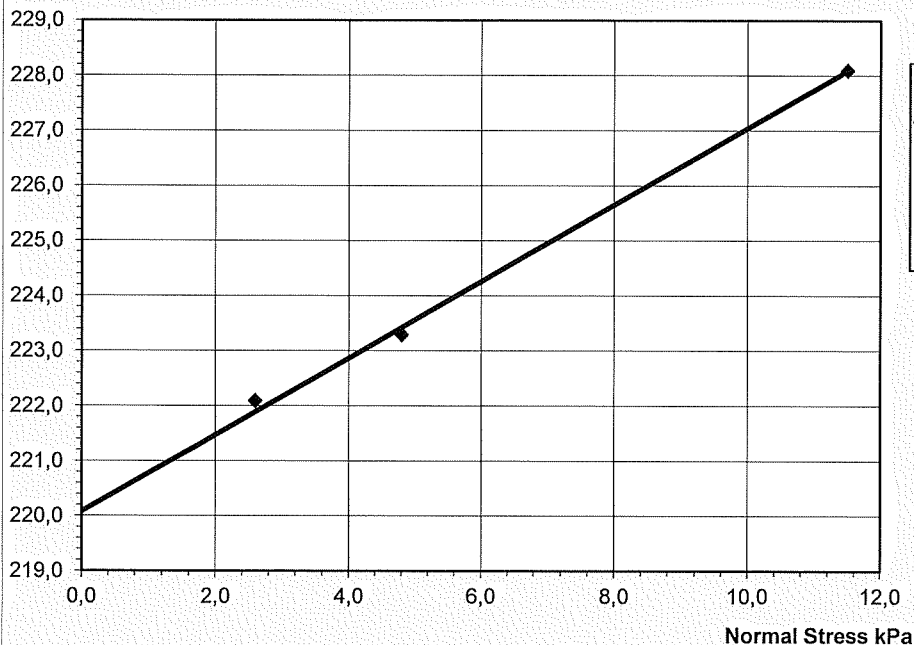
Reference	A	B	C
Applied Normal Stress	2,6 kPa	4,8 kPa	11,5 kPa
Peak Strength	222,1 kPa	223,3 kPa	228,1 kPa
Residual Shear Stress			
Rate(s) of Shear	Stage 1:	Stage 1:	Stage 1:
Displacement	0,9429mm/min	0,9326mm/min	0,9227mm/min
Final Height	19,99 mm	19,97 mm	19,94 mm
Bulk Density	2,234 Mg/m ³	2,252 Mg/m ³	2,244 Mg/m ³
Dry Density	2,122 Mg/m ³	2,141 Mg/m ³	2,131 Mg/m ³
Moisture Content	5,3 %	5,2 %	5,3 %
Number of Traverses	1	1	1

Condition of sample: Remoulded

Area : 60 mm x 60 mm

Maximum Shear Stress vs Normal Stress

Peak Shear Stress kPa



Peak ◆
Angle of Shear Resistance
34.85 Degrees
Cohesion
220.08 kPa

Shear Strenght by direct shear (small shearbox) CEN ISO/TS 17892-10: 2004



Client:ENERGYINTEL SERVICES LTD

Project: PLOT 316

Site: SHA

Sampler : GEOINVEST LTD

Trial Pit No. : 4

Depth : 1,00-1,50m

Description of material: Soil Cover

Date of test completion: 17/09/2025

Operator : Ch.P.

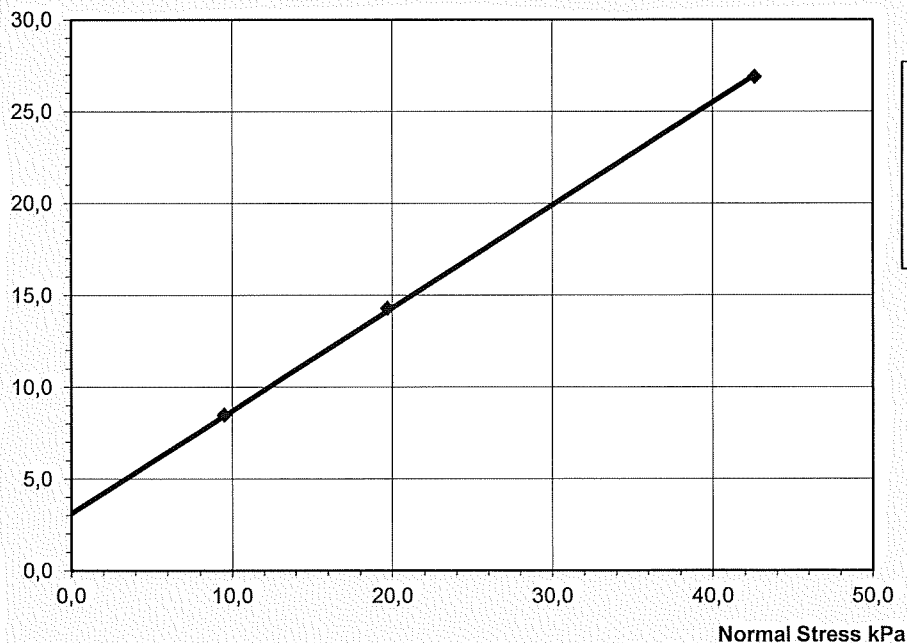
Reference	A	B	C
Applied Normal Stress	9,5 kPa	19,7 kPa	42,6 kPa
Peak Strength	8,5 kPa	14,3 kPa	26,9 kPa
Residual Shear Stress			
Rate(s) of Shear Displacement	Stage 1: 0,9805mm/min	Stage 1: 0,97435mm/min	Stage 1: 0,9525mm/min
Final Height	19,99 mm	19,97 mm	19,95 mm
Bulk Density	1,598 Mg/m ³	1,550 Mg/m ³	1,575 Mg/m ³
Dry Density	1,465 Mg/m ³	1,419 Mg/m ³	1,442 Mg/m ³
Moisture Content	9,1 %	9,2 %	9,2 %
Number of Traverses	1	1	1

Condition of sample: Remoulded

Area : 60 mm x 60 mm

Maximum Shear Stress vs Normal Stress

Peak Shear Stress kPa



Peak

Angle of Shear Resistance
29,24 Degrees

Cohesion
3.12 kPa

Shear Strength by direct shear (small shearbox) CEN ISO/TS 17892-10: 2004



Client: ENERGYINTEL SERVICES LTD
Project: PLOT 316
Site: SHA
Sampler : GEOINVEST LTD

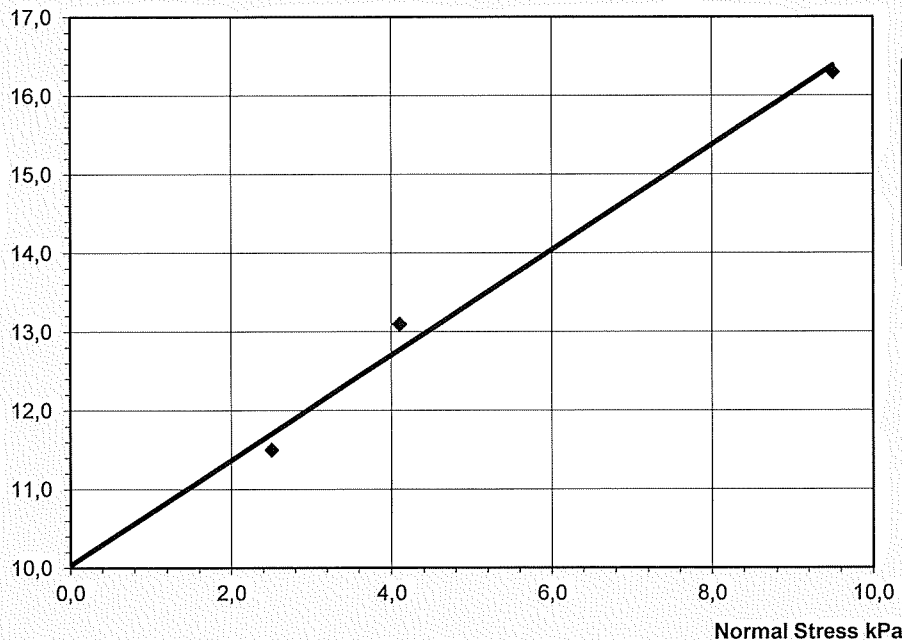
Trial Pit No. : 10
Depth : 0,00-1,00m
Description of material: Soil Cover
Date of test completion : 18/09/2025
Operator : Chr.I.

Reference	A	B	C
Applied Normal Stress	2,5 kPa	4,1 kPa	9,5 kPa
Peak Strength	11,5 kPa	13,1 kPa	16,3 kPa
Residual Shear Stress			
Rate(s) of Shear Displacement	Stage 1: 0,9392mm/min	Stage 1: 0,9512mm/min	Stage 1: 0,9845mm/min
Final Height	19,93 mm	19,95 mm	19,96 mm
Bulk Density	1,862 Mg/m ³	1,915 Mg/m ³	1,890 Mg/m ³
Dry Density	1,727 Mg/m ³	1,778 Mg/m ³	1,753 Mg/m ³
Moisture Content	7,8 %	7,7 %	7,8 %
Number of Traverses	1	1	1

Condition of sample: Remoulded
Area : 60 mm x 60 mm

Maximum Shear Stress vs Normal Stress

Peak Shear Stress kPa



Peak ◆
Angle of Shear Resistance
33.77 Degrees
Cohesion
10.03 kPa